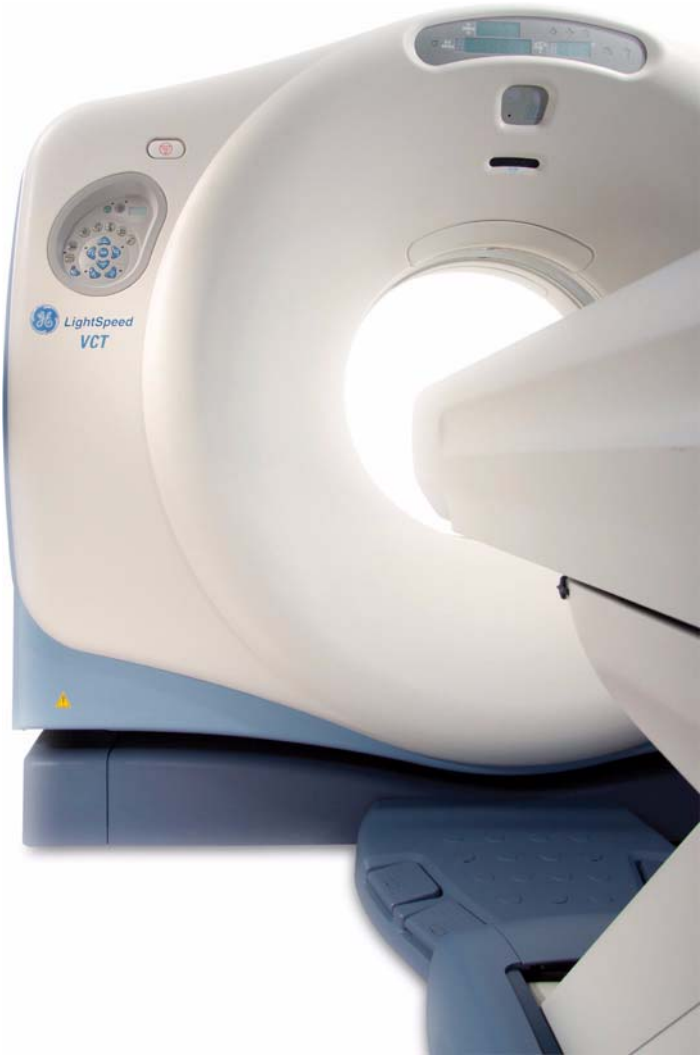


LightSpeed 7.X
Functional Interconnects & Block Diagrams



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IMPORTANT PRECAUTIONS

LANGUAGE		
WARNING (EN)	- This Service Manual is available in English only. - If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.	ADVARSEL (DA) - Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
Предупреждение (BG)	- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК. - АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД. - НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА. - НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА - ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.	WAARSCHUWING (NL) - Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
警告 (ZH-CN)	- 本维修手册仅提供英文版本。 - 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。	HOIATUS (ET) - Käesolev teenindusjuhend on saadaval ainult inglise keeles. - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VÝSTRAHA (CS)	- Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.	VAROITUS (FI) - Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
		ATTENTION (FR) - Ce manuel de service n'est disponible qu'en anglais. - Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNING (DE)	• Diese Serviceanleitung existiert nur in Englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.	警告 (JA)	• このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	• Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. • Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εαν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. • Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.	경고 (KO)	• 본 서비스 지침서는 영어로만 이용하실 수 있습니다 . • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다 . • 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 . • 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다 .
FIGYELMEZTETÉS (HU)	• Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. • Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.	BRDINJUMS (LV)	• Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu. • Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.
AÐVÖRUN (IS)	• Þessi þjónustuhandbók er eingöngu fánleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. • Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.	ĮSPĖJIMAS (LT)	• Šis eksploatavimo vadovas yra prieinamas tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė. • Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote. • Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.
AVVERTENZA (IT)	• Il presente manuale di manutenzione è disponibile soltanto in inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.	ADVARSEL (NO)	• Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE (PL)	- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. - Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. - Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego. - Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.	ATENCION (ES)	- Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
ATENÇÃO (PT)	- Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.	VARNING (SV)	- Den här servicehandboken finns bara tillgänglig på engelska. - Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
ATENȚIE (RO)	- Acest manual de service este disponibil numai în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.	DIKKAT (TR)	- Bu servis kilavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyeni bu kilavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.
ОСТОРОЖНО! (RU)	- Данное руководство по обслуживанию предлагается только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.		
UPOZORNENIE (SK)	- Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.		

DAMAGE IN TRANSPORTATION

Check for damage to property that may have occurred at the site during delivery, such as damage to floors, door frames or walls.

All packages should be closely examined at time of delivery. If damage is apparent, have the notation *Damage in Shipment* written on **all** copies of the freight or express bill before delivery is accepted or signed for by a GE Healthcare representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

Call 1-800-548-3366, and use option 8.

Fill out a report on <http://egems.med.ge.com/edq/home.jsp>

Contact the local service coordinator for more information about this process.

Rev. September 17, 2006

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment, if not properly used, may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before attempting to place this equipment in operation. The General Electric Company, GE Healthcare Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, GE Healthcare Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

IMPORTANT...RADIOACTIVE MATERIAL HANDLING

Only employees formally trained in radioactive materials handling and this equipment are authorized by the GE Healthcare Radiation Safety Officer to use radioactive materials to service this equipment.

GE Healthcare is required to notify the applicable U.S. state agency **PRIOR** to any source service event involving pin source handling. See NUC/PET radioactive material guides for specific instruction or contact your EHS Specialist.

A radiation survey must be performed when a pin source has been removed and replaced. See Radiation Survey Form Instructions or contact your EHS Specialist.

Rev 2 (July 21, 2005)

LITHIUM BATTERY CAUTIONARY STATEMENTS

Risk of Explosion

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer’s instructions.

Danger d’Explosion

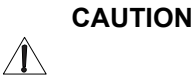
Il y a danger d’explosion s’il y a remplacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d’un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers: please contact the GE Healthcare Sales or Service representatives.

GE personnel: please use the GE Healthcare iTrak/PQR Process to report all omissions, errors, and defects in this publication.



Revision History

Rev	Date	Reason for change
1	3/09/06	Initial Release.
2	7/12/06	PQR 13071670 Chapter 10: Updated Detector Heater Control and Detector Plenum Fan Control drawings (included drawings for 2005 & 2006 styles)
3	11/28/06	Chapter 6: Updated VDAS/Detector Backplane Map; added HDAS/Detector Backplane Map. Chapter 10: Updated Detector Plenum Fan Control (2006 Style) drawing
4	11/17/07	Chapter 6: Updated HDAS/Detector Backplane Map. Chapter 6: Corrected Trigger Generation, 2399724 (DAS-to-JEDI wiring) Chapter 10: Replaced diagram for Detector Plenum Fan Control (2006 Style) with correct diagram.

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Chapter 11

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Section 1.0 — System E-Stop and Power Interlocks - 2399742

Section 2.0 — HVDC/120V Control - 2399744

Section 3.0 — Gantry Power Distribution - 5128042

Section 4.0 — PDU - Gantry 120 VAC Control - 5128048

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Chapter 1

Introduction

Section 1.0 — Drawing Legibility

The content of this manual is designed to be viewed and used in its electronic form on a computer monitor. FCT Service Engineering supports the electronic version of this manual only. By design, when viewed in its printed form, drawing sizes and font characteristics may not be legible. Locally printed versions of this manual are uncontrolled documents and cannot be used to service the products they support, without referring to the electronic version.

Section 2.0 — System Interconnects (5112889)

The Stationary and Rotating Gantry Table Interconnect drawings, (5112889) found on [page 19](#) and [page 24](#), are optimized for electronic review. They are the foundation drawings for all the individual drawings found in this manual. Refer to these drawings in their electronic form for electronic magnification, for additional interconnect details for this product.

Section 1.0 — VCT System Block Diagram

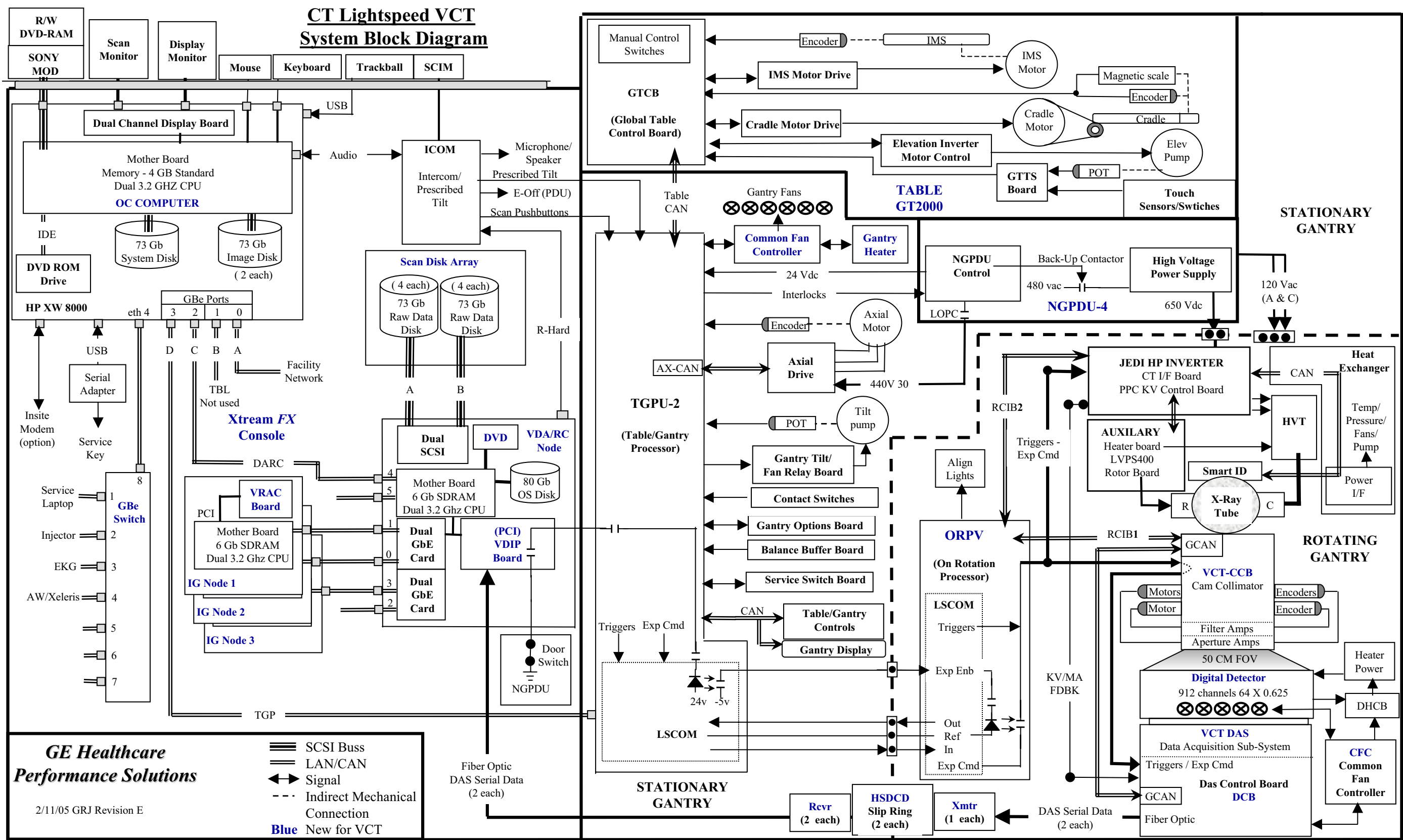


Figure 2-1 VCT System Block Diagram, Rev E

Section 2.0 — Gantry Stationary Block Diagram - 5128051

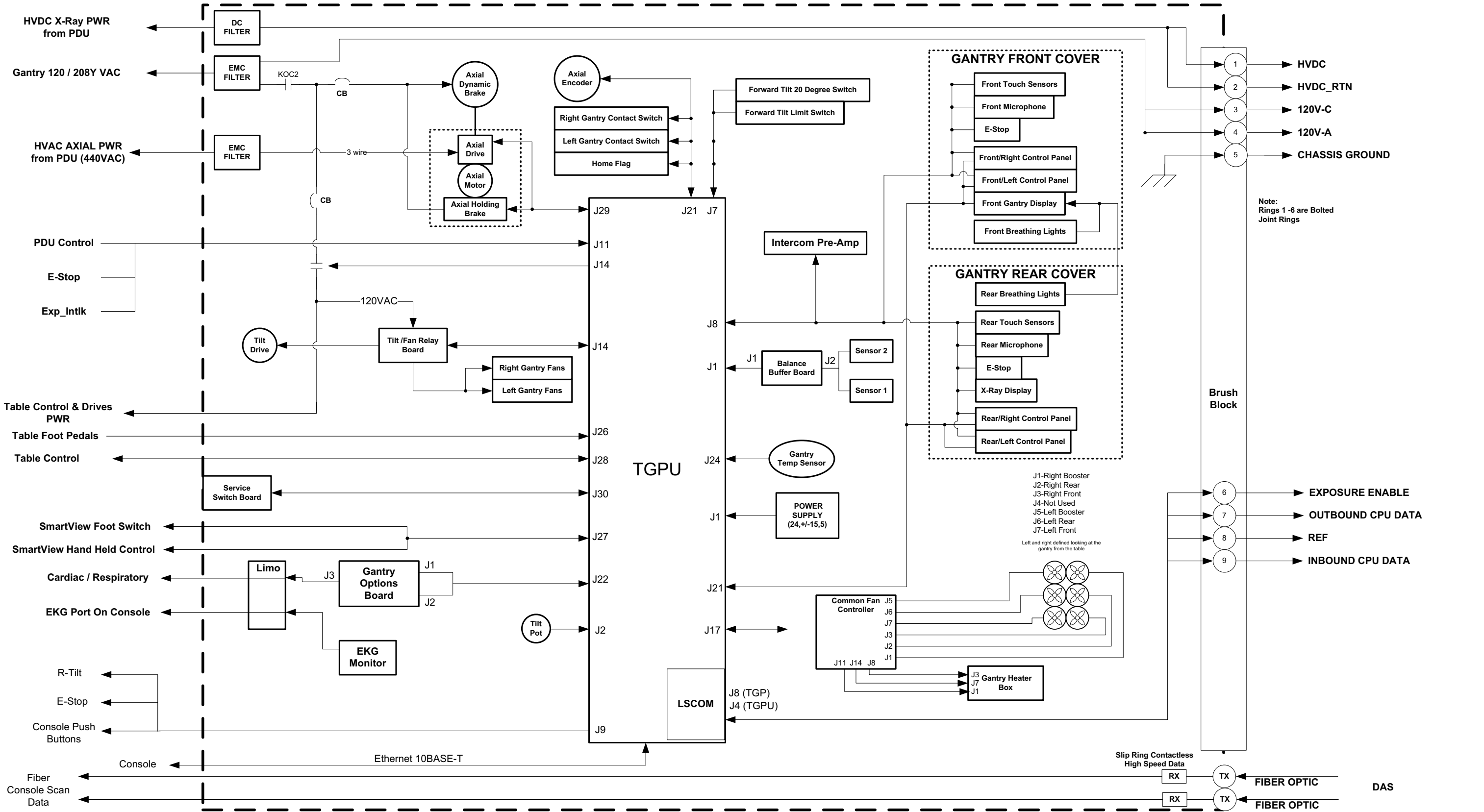
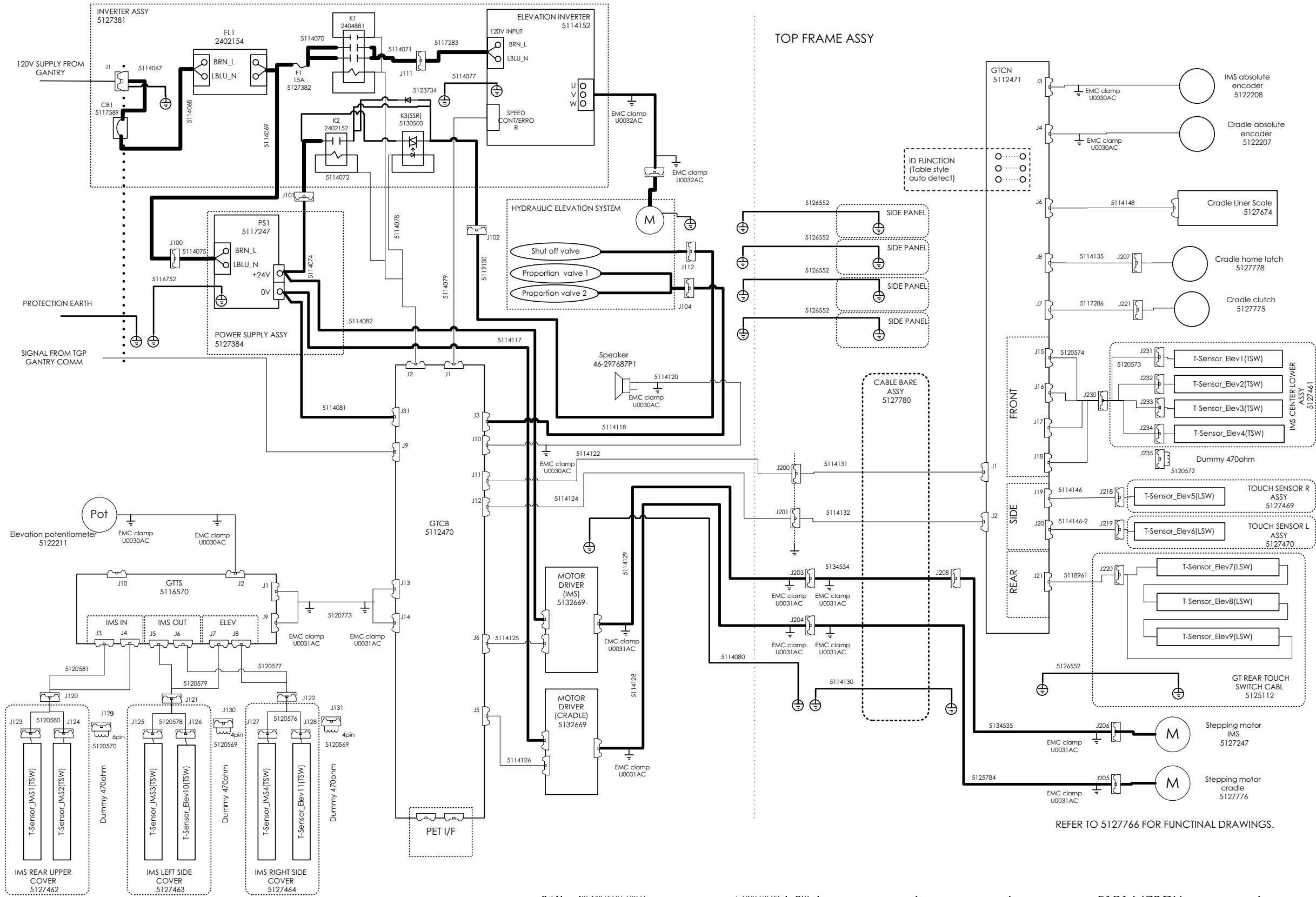


Figure 2-2 5128051 Sheet 1, Revised 10/05/05

Section 3.0 — Table Block Diagram - 5121647



REFER TO 5127766 FOR FUNCTIONAL DRAWINGS.

Figure 2-3 5121647 Sheet 1, Rev. 1

Section 4.0 — Console Block Diagram - 5115335

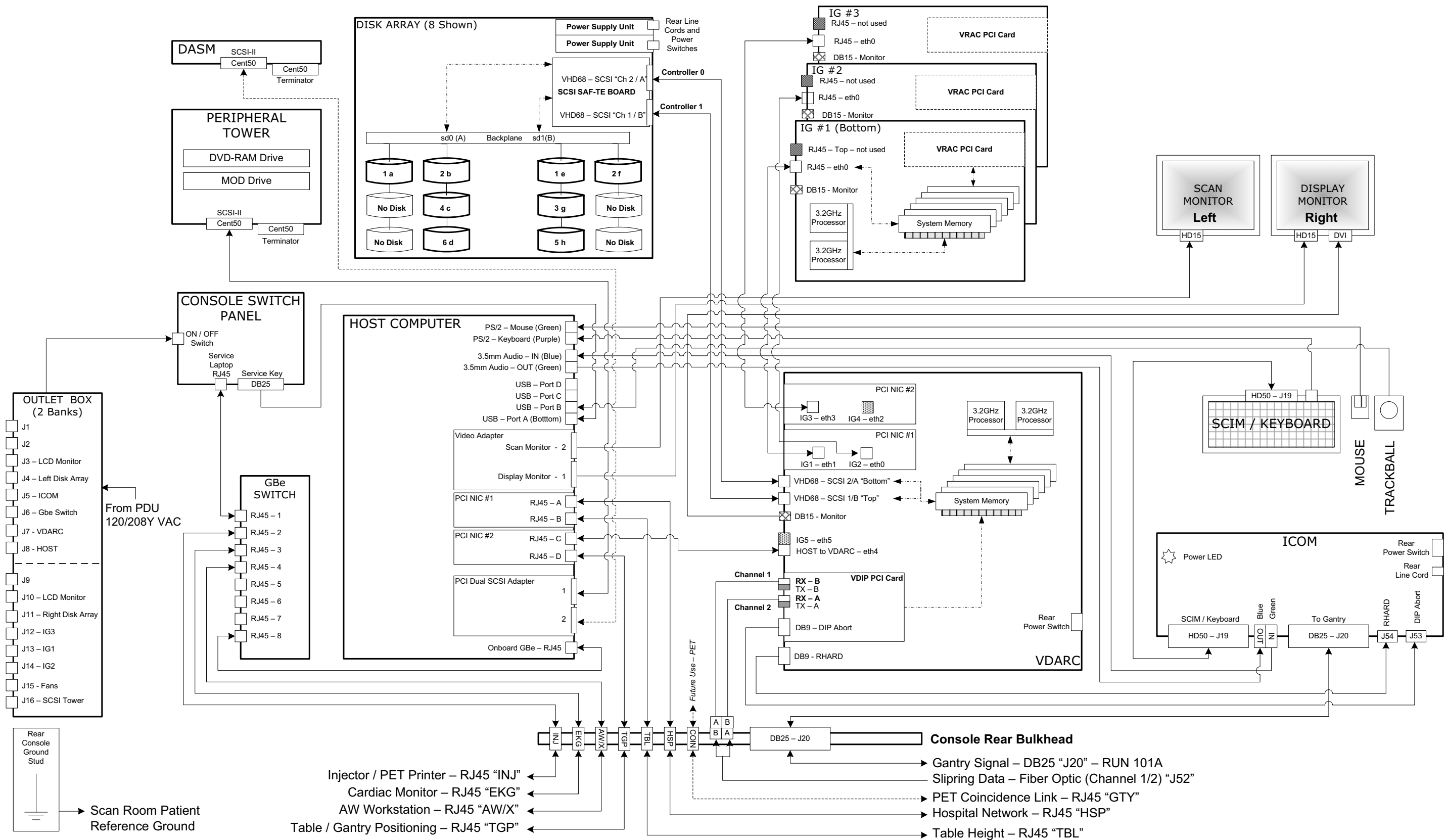


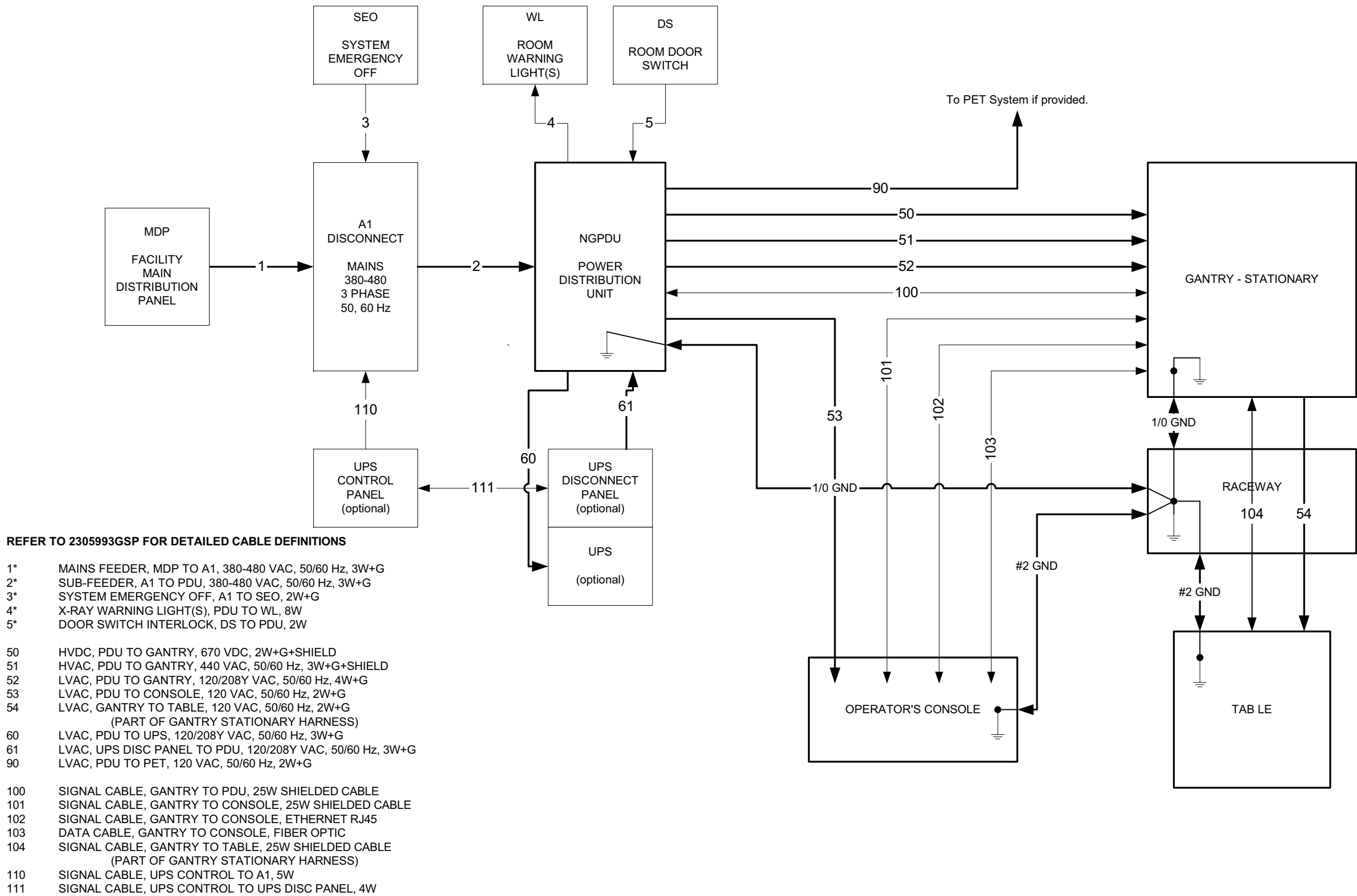
Figure 2-4 5115335 Sheet 1, Rev 2

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Chapter 3

Cable Interconnects

Section 1.0 — System Cable Interconnect - 2305993

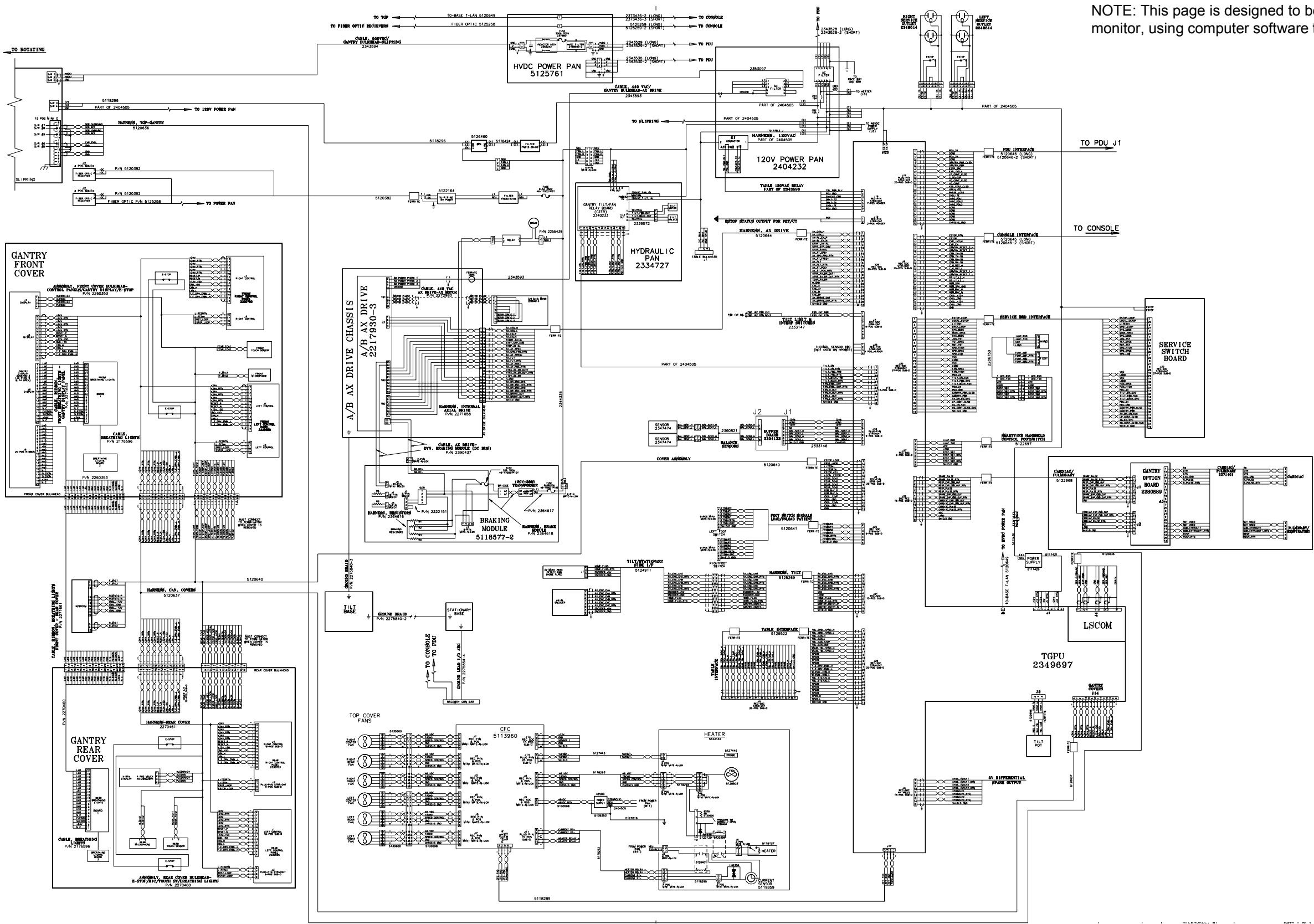


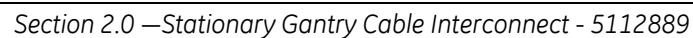
* NOTE: Runs 1, 2, 3, 4 & 5 are customer supplied. All others are supplied by GEMS.

Figure 3-1 2305993 Sheet 1, Rev D

Section 2.0 — Stationary Gantry Cable Interconnect - 5112889

NOTE: This page is designed to be viewed exclusively on a computer monitor, using computer software to zoom on the drawing (for legibility).





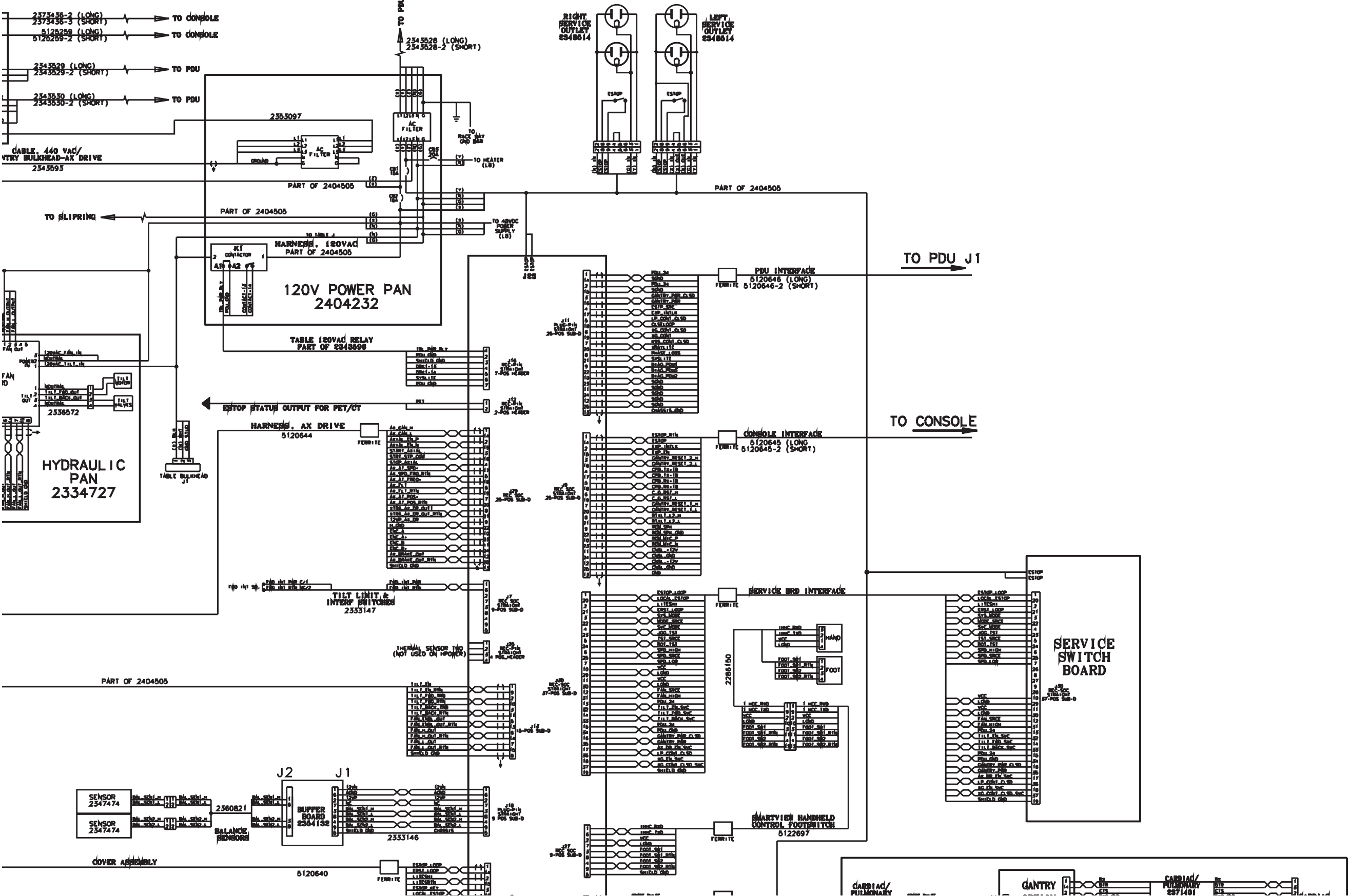


Figure 3-4 5112889 Sheet 1, Rev 6 - Upper Right Quadrant

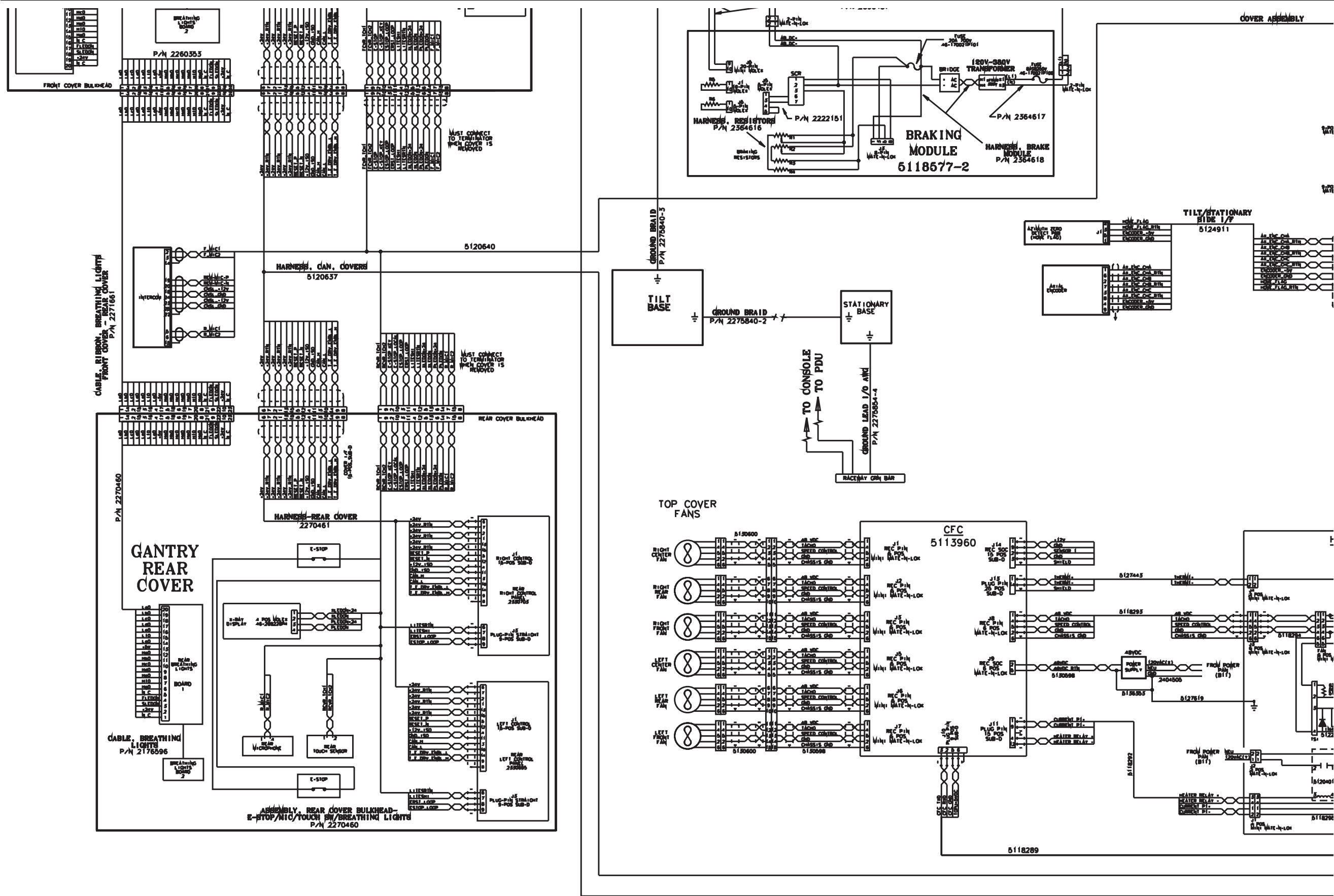


Figure 3-5 5112889 Sheet 1, Rev 6 - Lower Left Quadrant

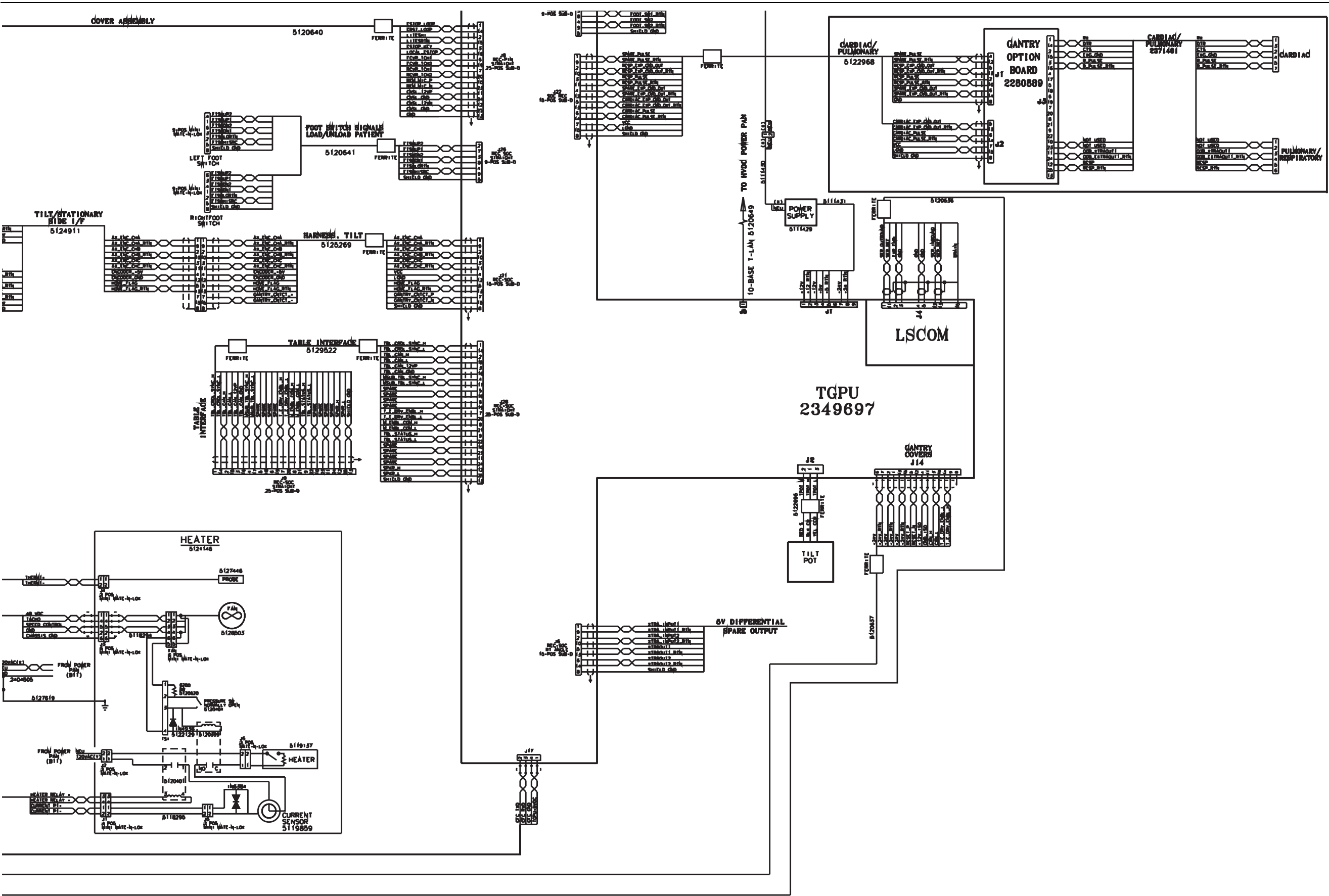


Figure 3-6 5112889 Sheet 1, Rev 6 - Lower Right Quadrant

Section 3.0 — Rotating Gantry Cable Interconnect - 5112889

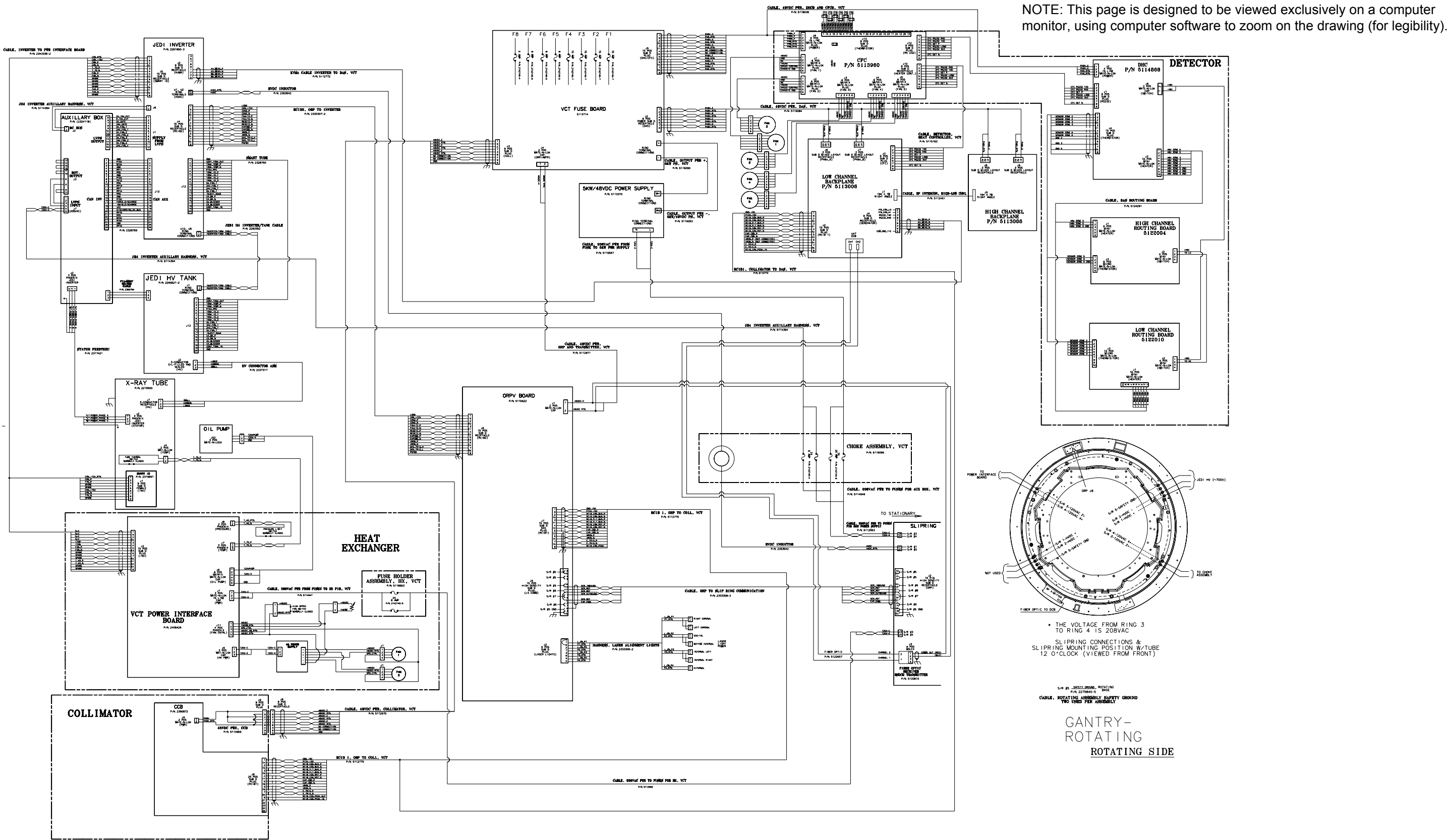


Figure 3-7 5112889 Sheet 2, Rev. 6

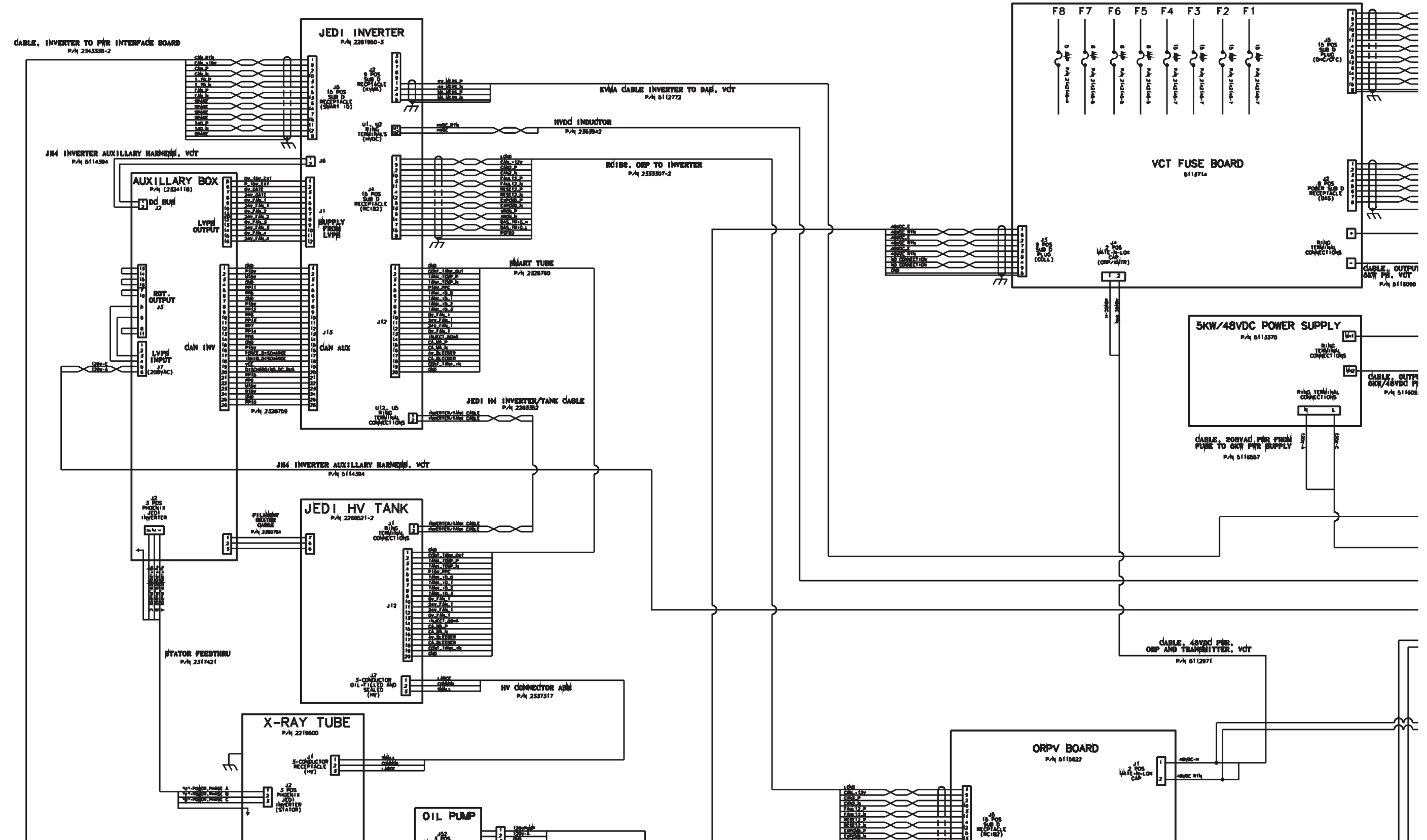


Figure 3-8 5112889 Sheet 2, Rev. 6 - Upper Left Quadrant

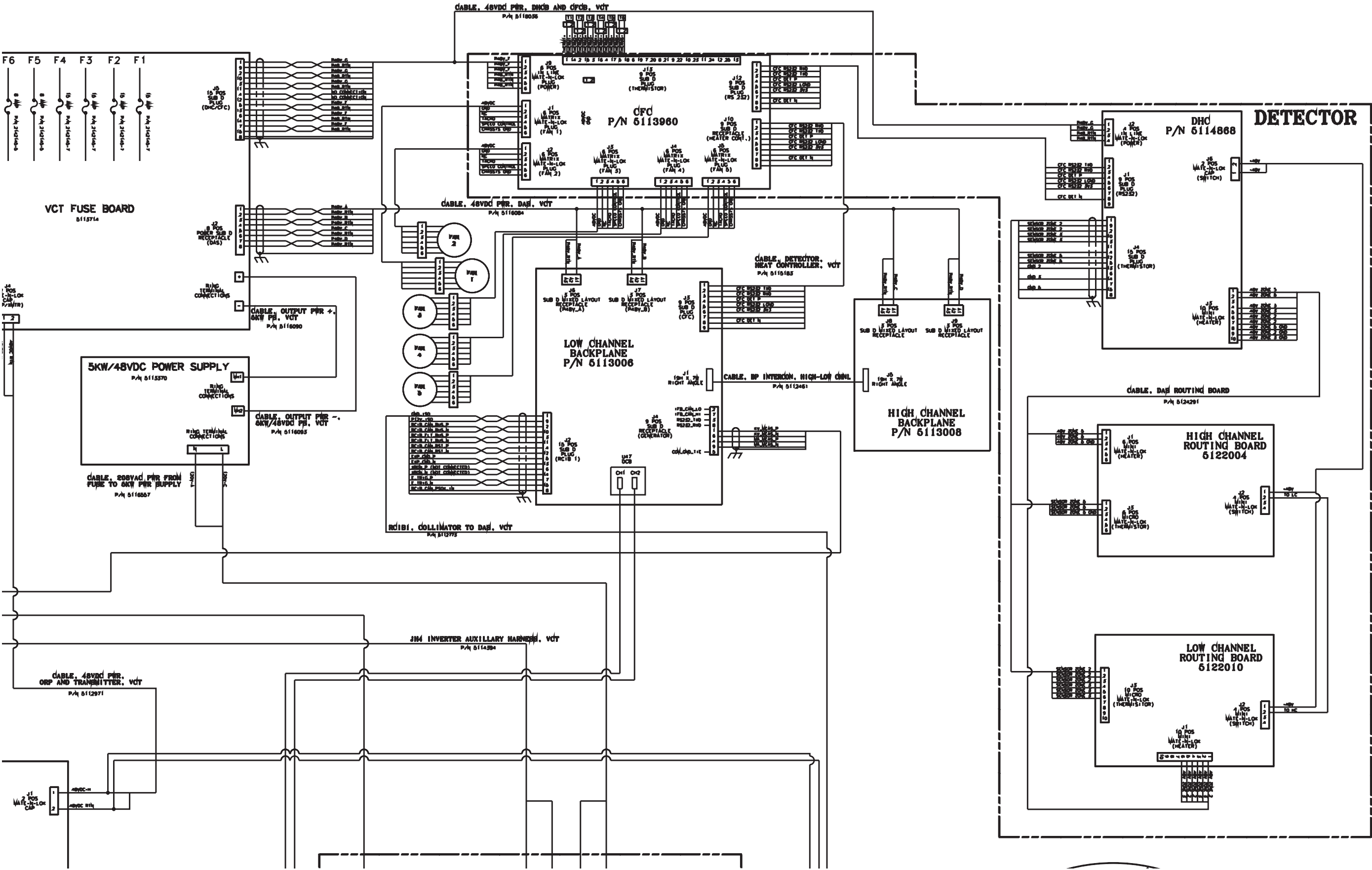


Figure 3-9 5112889 Sheet 2, Rev. 6 - Upper Right Quadrant

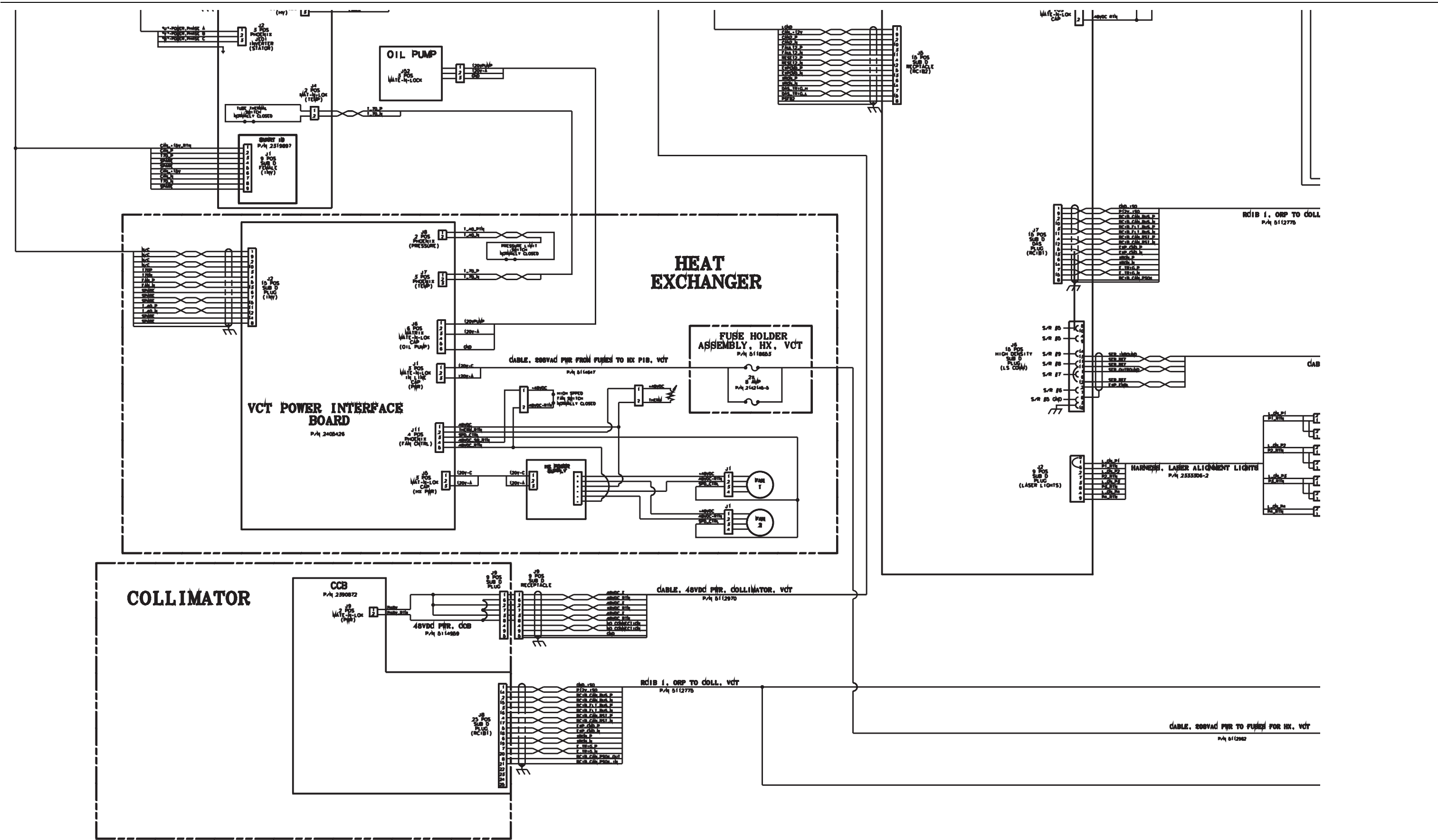
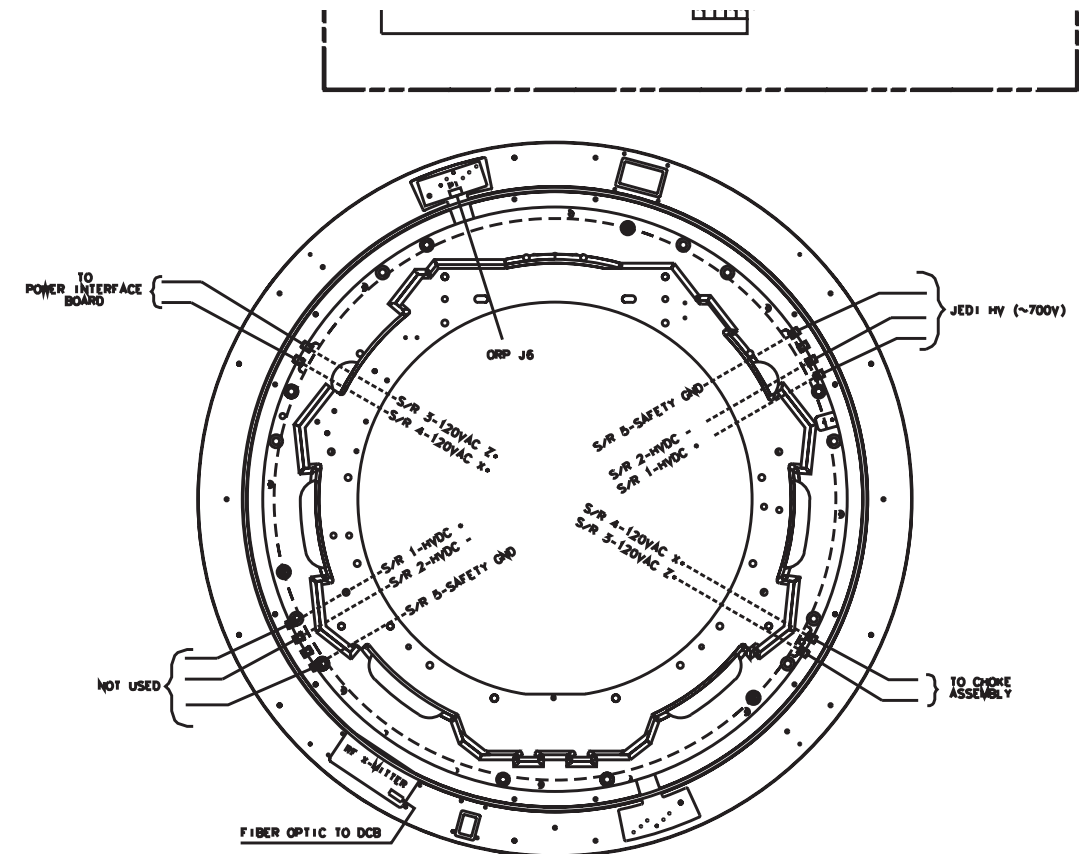


Figure 3-10 5112889 Sheet 2, Rev. 6 - Lower Left Quadrant



S/R #5 SAFETY GROUND ROTATING
P/N 2275840-5 BASE

CABLE, ROTATING ASSEMBLY SAFETY GROUND
TWO USED PER ASSEMBLY

GANTRY- ROTATING

Page 28 Section 3.0 – Rotating Gantry Cable Interconnect - 5112889

Chapter 4

System Communications

Section 1.0 — Slip Ring Communications Interface - 5128026

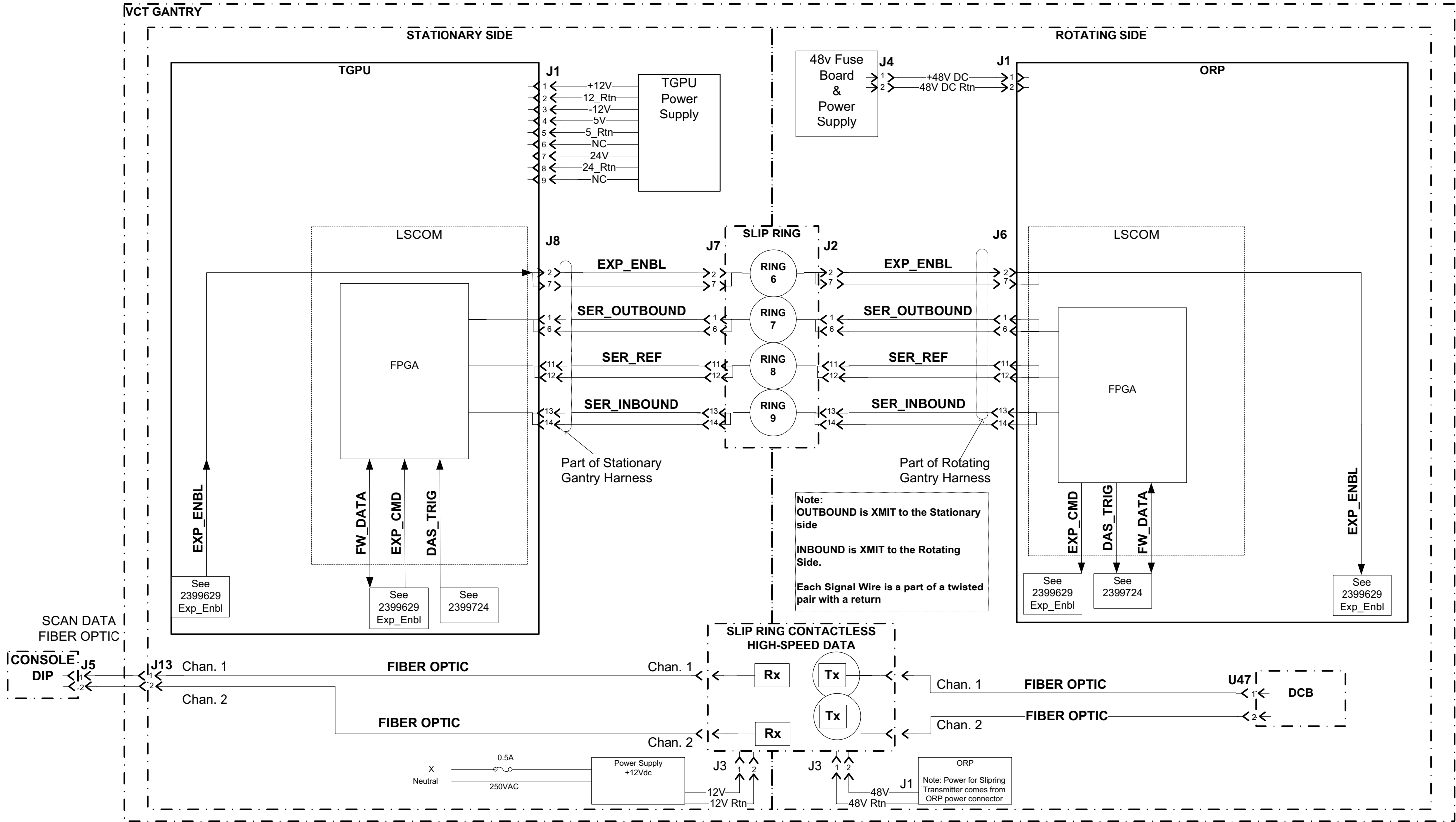


Figure 4-1 5128026 Sheet 1, Rev 2

Section 2.0 — Rotating Controller Interface Bus (RCIB) - 5128028

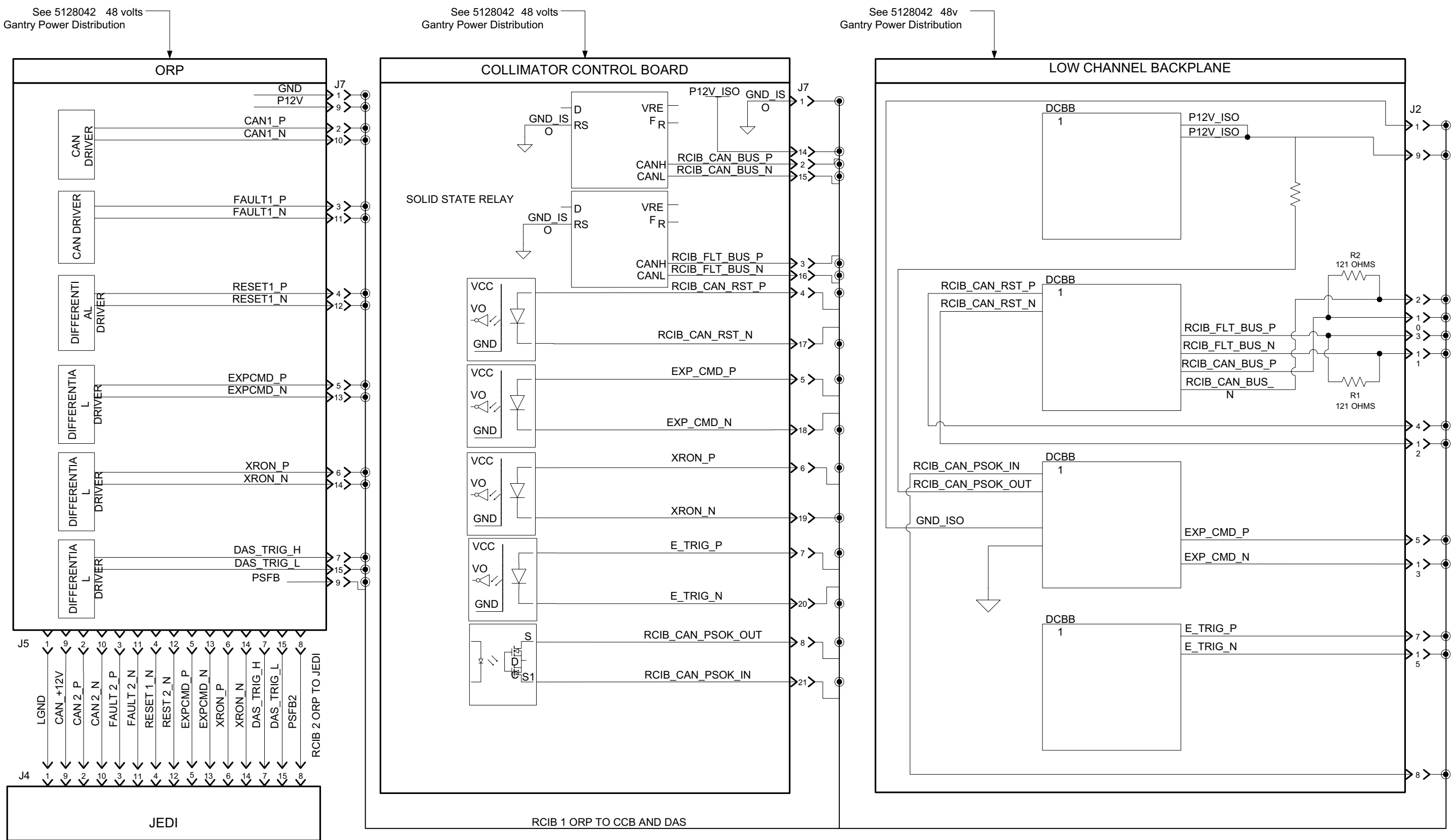


Figure 4-2 5128028 Sheet 1, Rev 2

Section 3.0 — Rotating Controller Interface Bus (RCIB) - Additional Detail

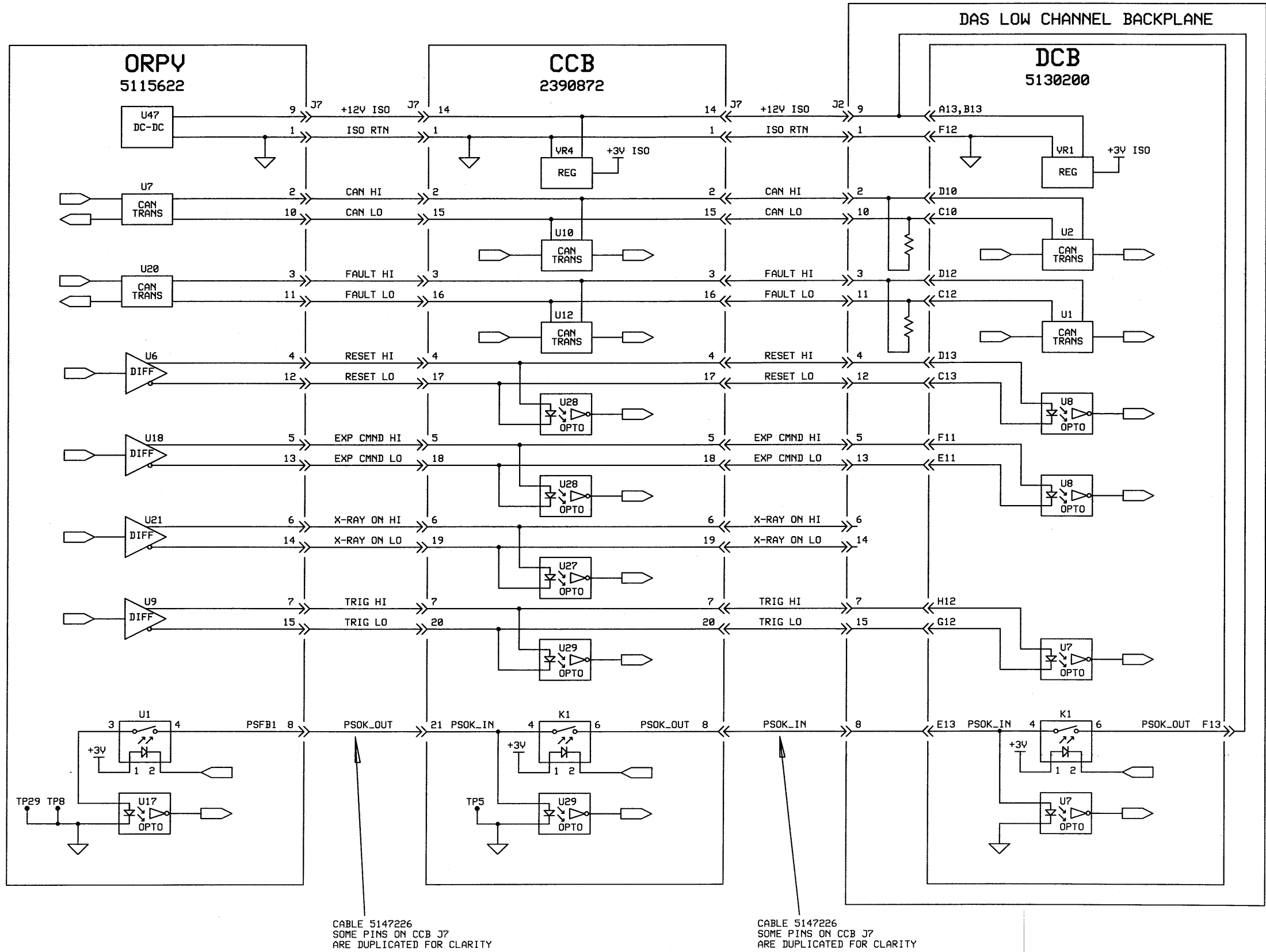


Figure 4-3 VCT RCIB Block Diagram

Chapter 5

X-Ray Generation

Section 1.0 – Tube Fans / Pump / SmartID Interface - 5128030

Section 2.0 — JEDI Interface - kV Loop - 2399625

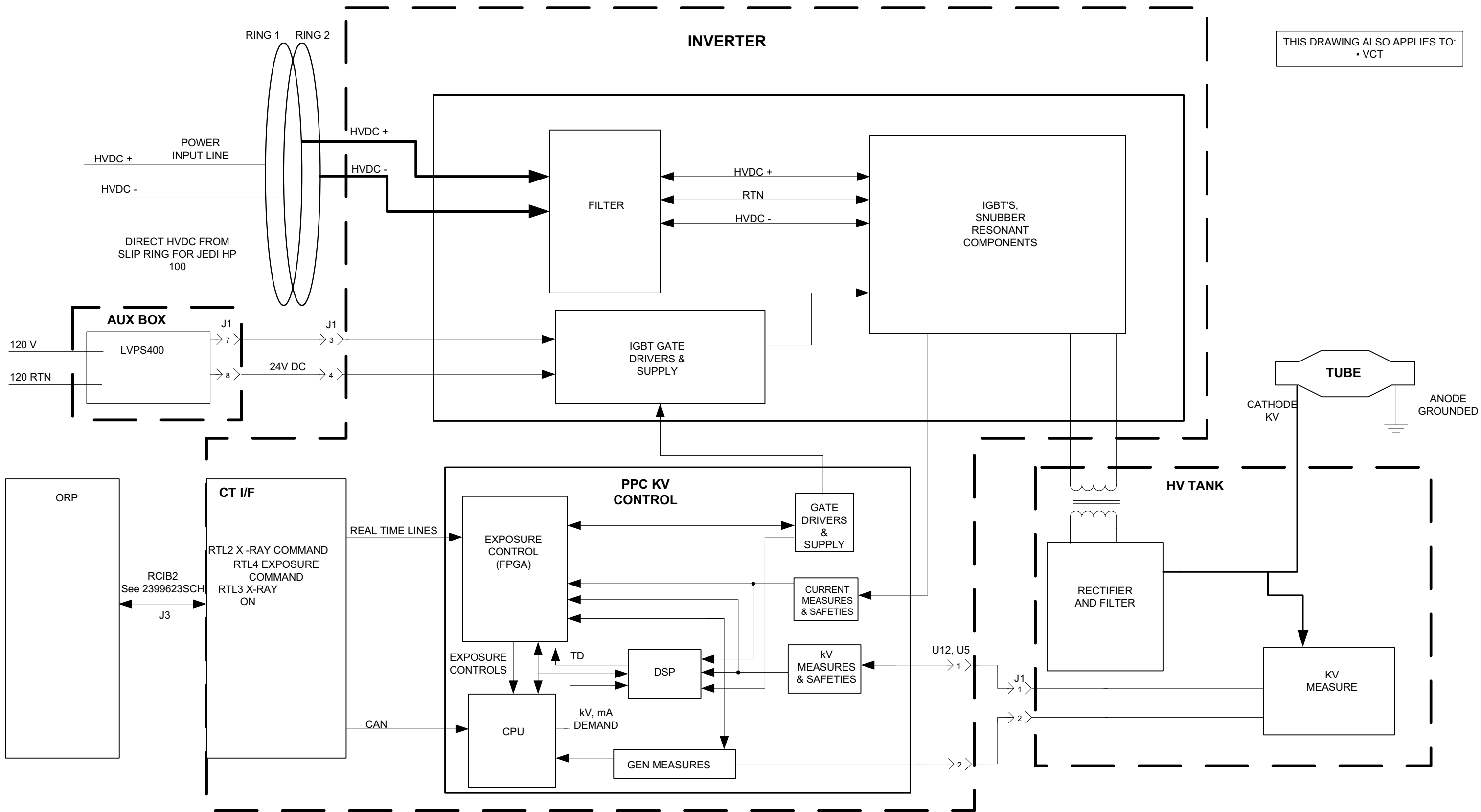


Figure 5-2 2399625 Sheet 1, Rev. 2

Section 3.0 — JEDI Interface - mA Loop - 2399628

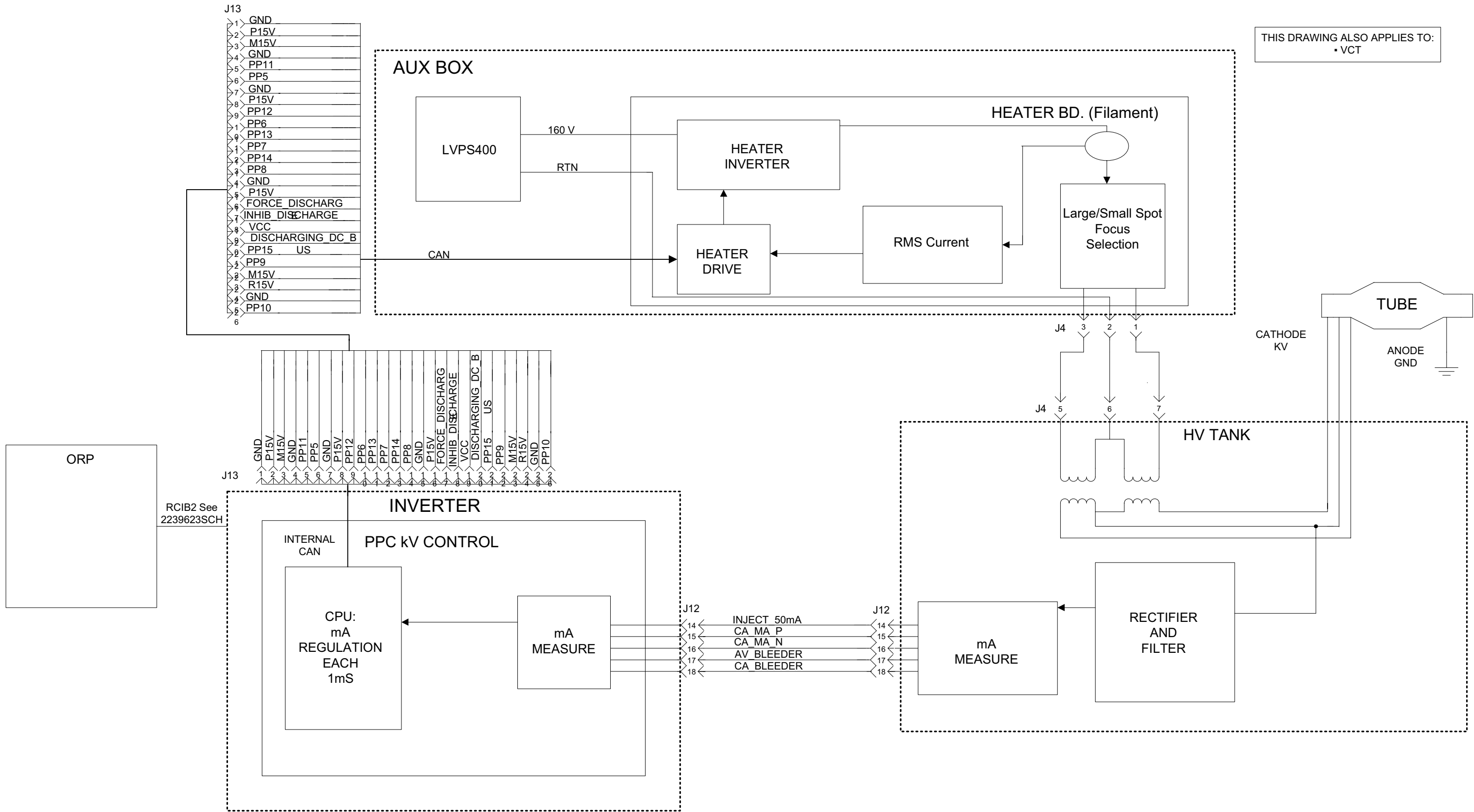


Figure 5-3 2399628 Sheet 1, Rev. 2

Section 4.0 — X-Ray Collimation & Filtration - 5128031

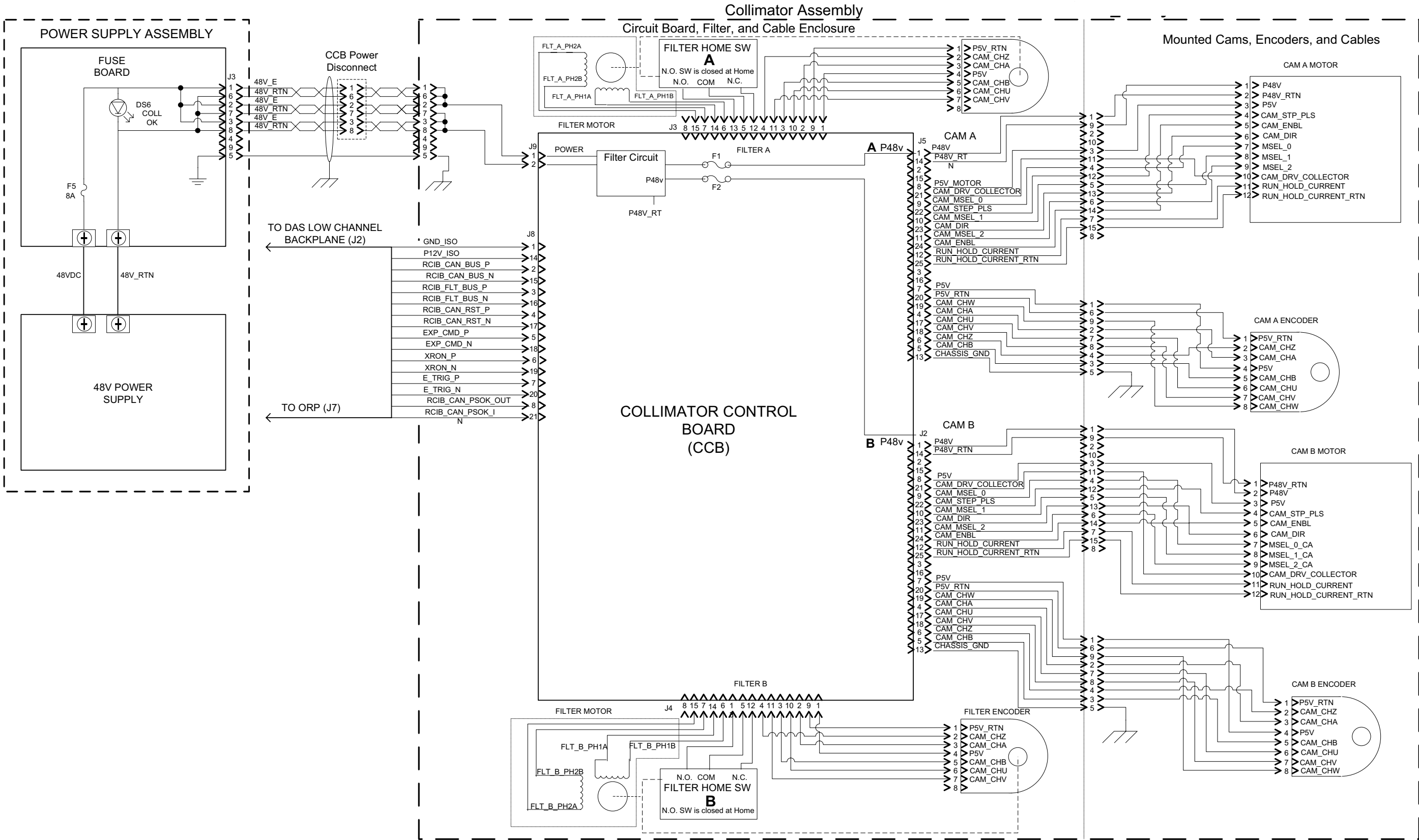


Figure 5-4 5128031 Sheet 1, Rev. 2

Section 5.0 — Exposure Enable and Final Command - 2399629

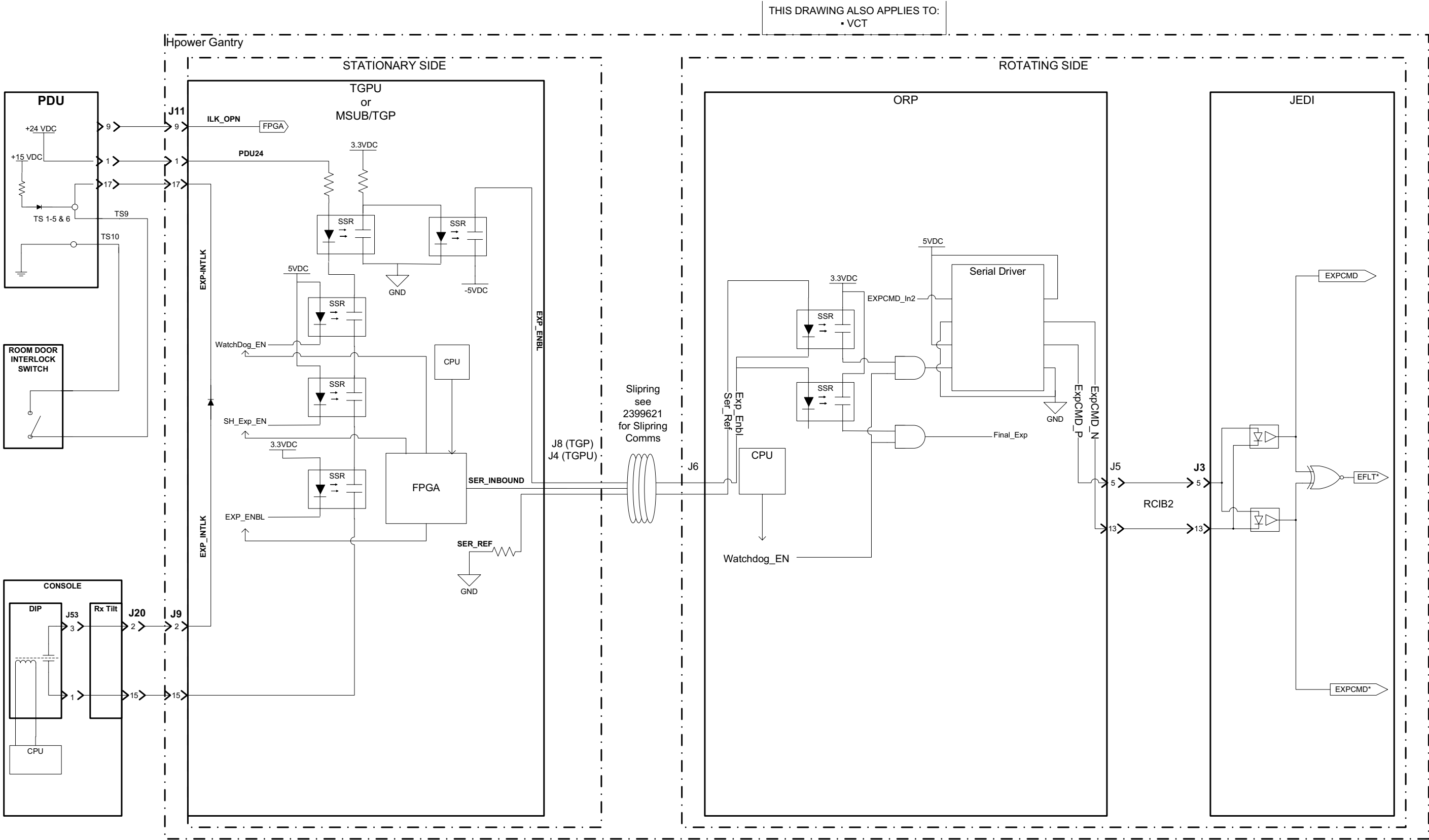


Figure 5-5 2399629 Sheet 1, Rev. 2

Section 6.0 — Tube Rotor Control - 2399630

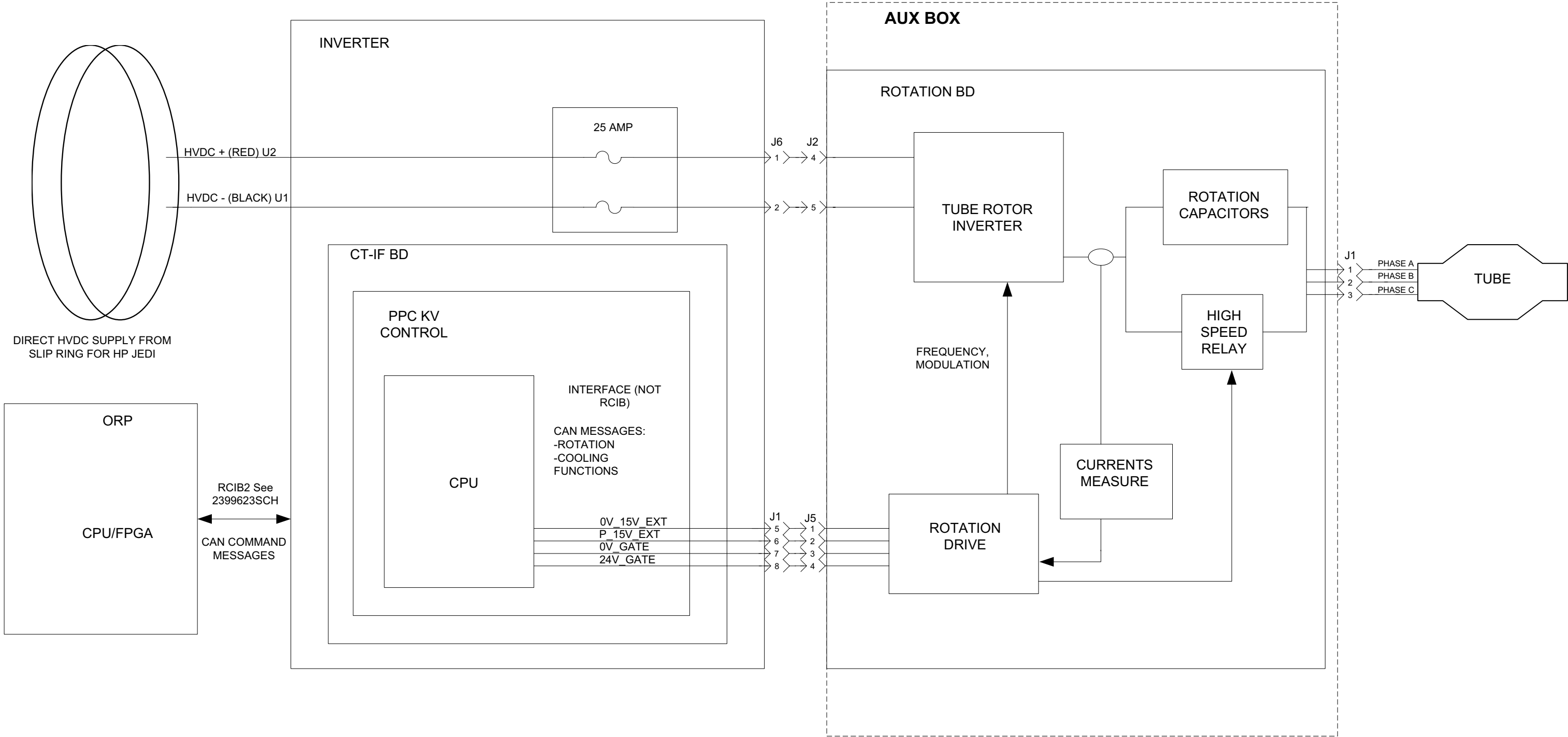
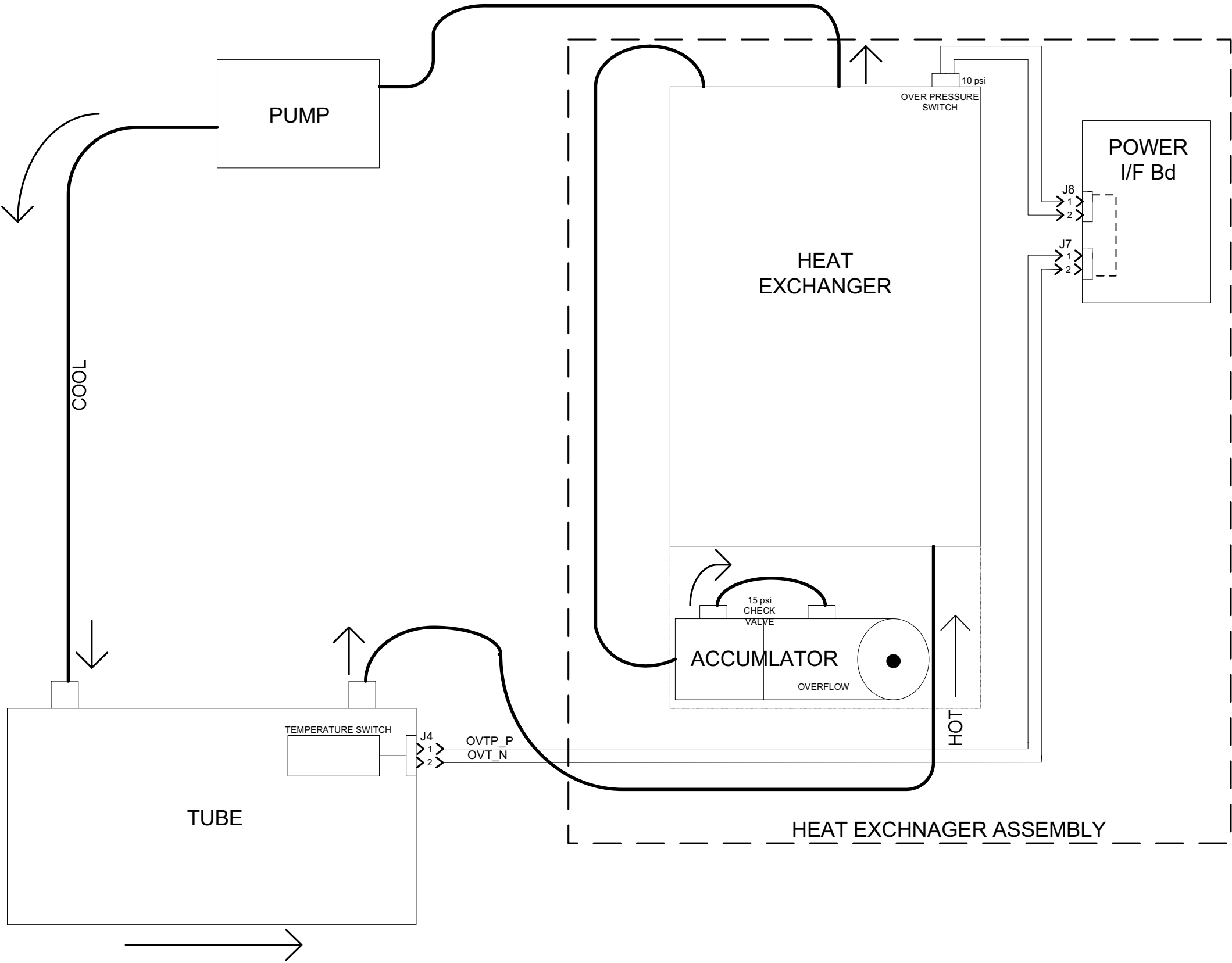


Figure 5-6 2399630 Sheet 1, Rev. 2

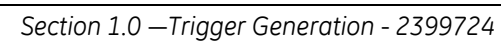
Section 7.0 — X-Ray Tube Coolant Flow



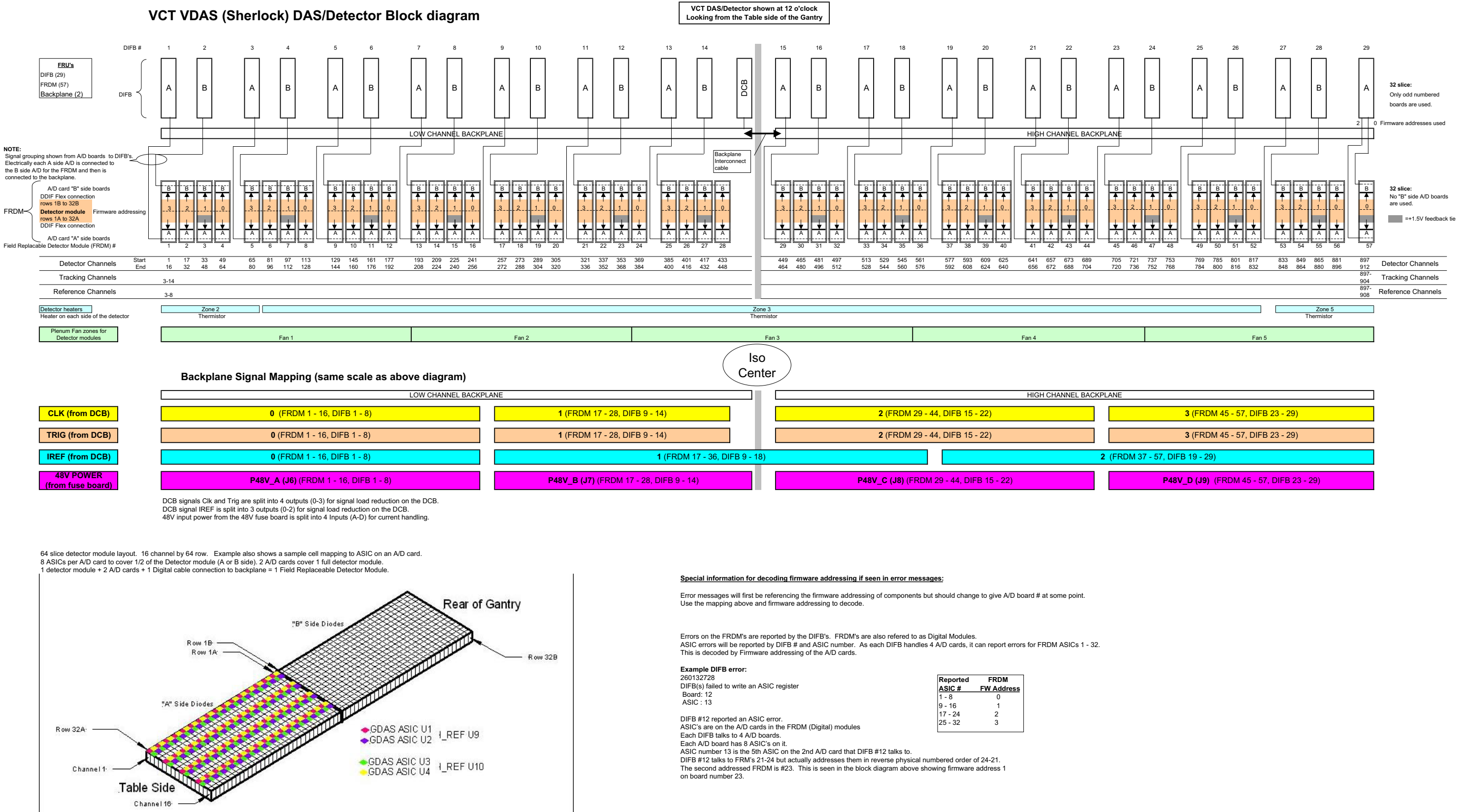
Chapter 6

Data Acquisition

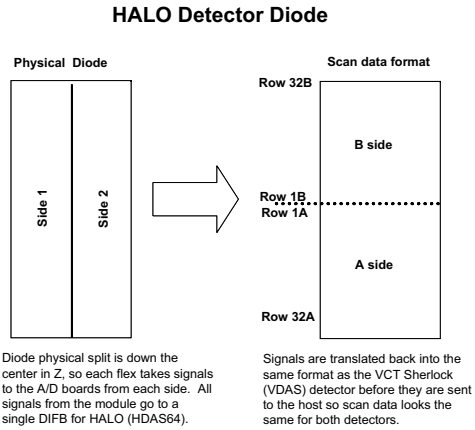
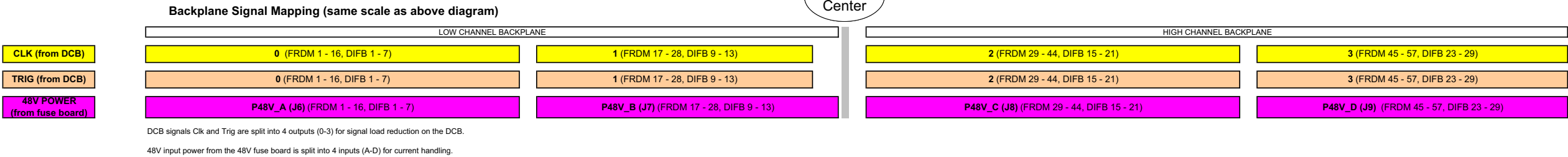
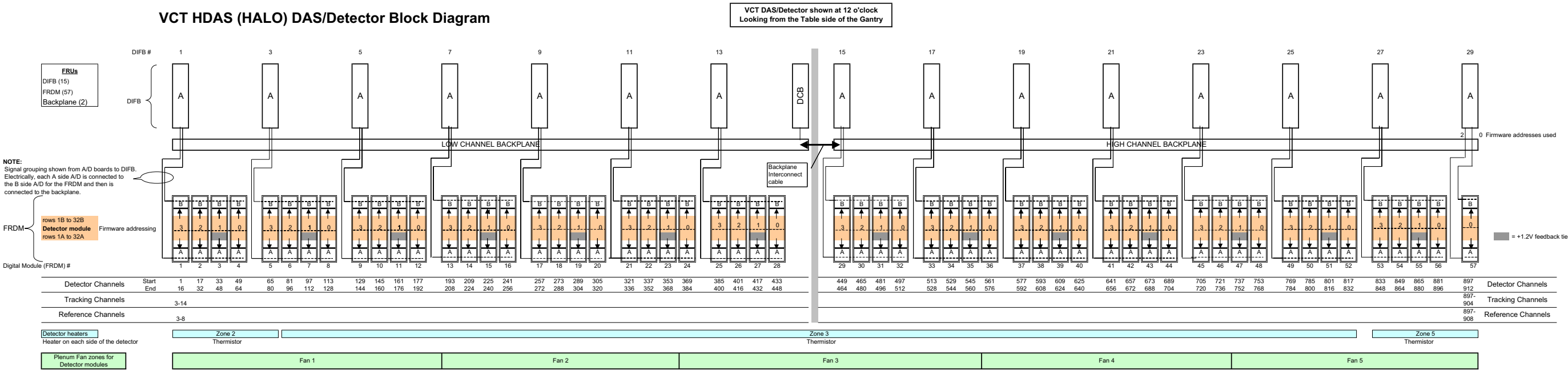
HP Trigger Generation



Section 2.0 — VDAS/Detector Backplane Map



Section 3.0 — HDAS/Detector Backplane Map



What does this impact?
Any issue will be either in the module or single DIFB. There is only a single DIFB for every 4 digital modules for HALO (HDAS). Still analyze the scan data the same way for both detectors (VDAS and HDAS). The data sorting occurs on the DIFBs then is sent to the DCB in the same format for both detectors. A flex failure would show up as an 8 ch x 64 row issue for HALO instead of the 16 ch x 32 row pattern for Sherlock.

Special information for decoding firmware addressing if seen in error messages:

Error messages should give digital module number, but if board number is not given, use the mapping above, firmware addressing, and the table and information below to decode.

Errors on the FRDMs are reported by the DIFBs. FRDMs are also referred to as Digital Modules. ASIC errors will be reported by DIFB # and ASIC number. As each DIFB handles 8 A/D cards, it can report errors for FRDM ASICs 0 - 63. This is decoded by firmware addressing of the A/D cards.

Example DIFB error:
260132728
DIFB(s) failed to write an ASIC register
Board: 11
ASIC: 13

Reported ASIC #	FRDM FW Address
0 - 15	0
16 - 31	1
32 - 47	2
48 - 63	3

DIFB #11 reported an ASIC error.
ASICs are on the A/D cards in the FRDM (Digital) modules.
Each DIFB talks to 8 A/D boards. Two A/Ds per digital module.
Each A/D board has 8 ASICs.
ASIC number 13 is the 14th ASIC that DIFB #11 talks to.
DIFB #11 talks to FRDMs 21-24 but actually addresses them in reverse physical numbered order of 24-21.
The first addressed FRDM is #24. This is seen in the block diagram above showing firmware address 0 on board number 24.

Figure 6-3 VCT HDAS/Detector Block Diagram, Rev 1)

Chapter 7

Patient Positioning

Section 1.0 — Alignment Lights - 2399726

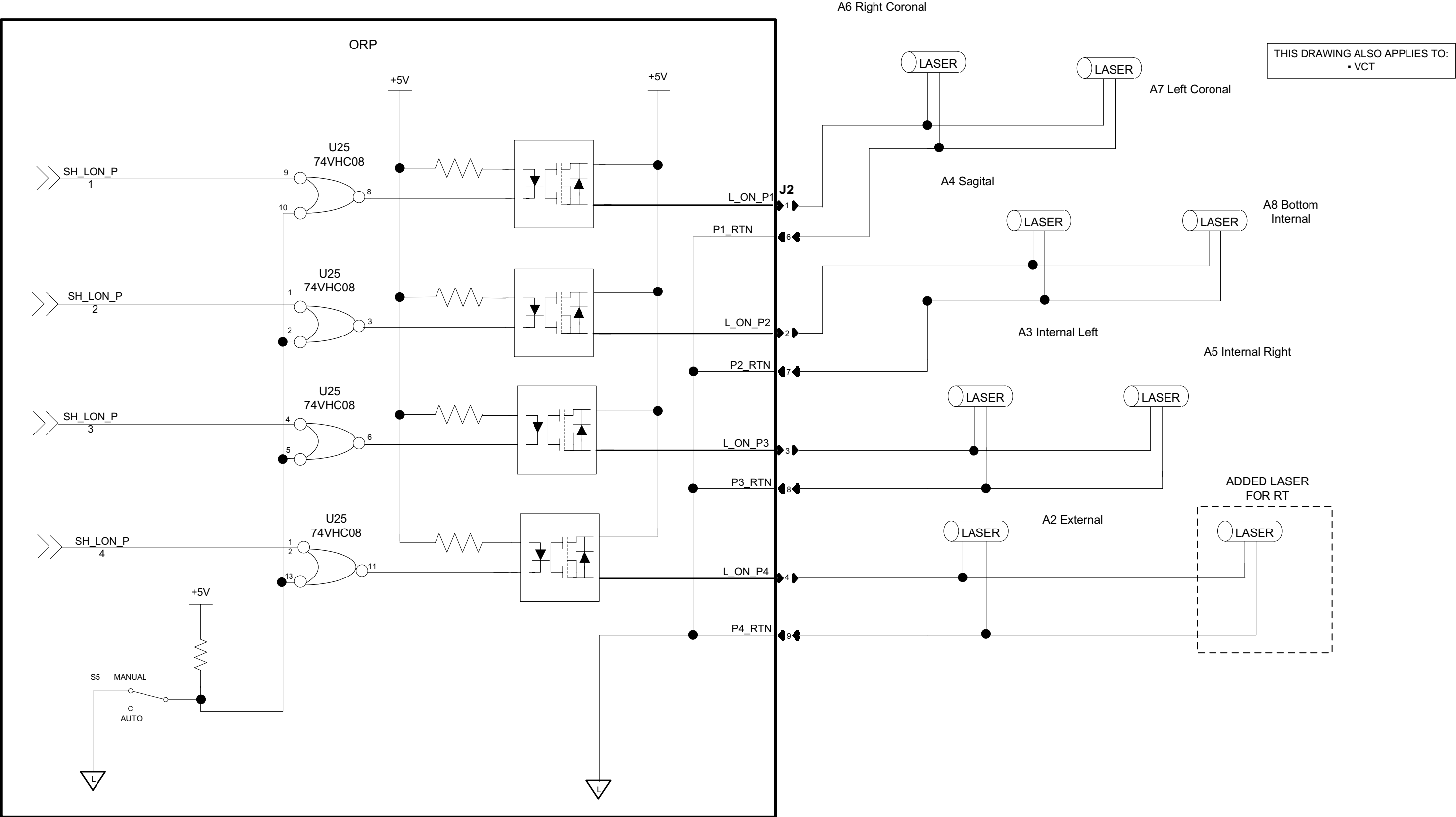


Figure 7-1 2399726 Sheet 1, Rev 2

Section 2.0 — Gantry Tilt, from Gantry, Console - 2399727

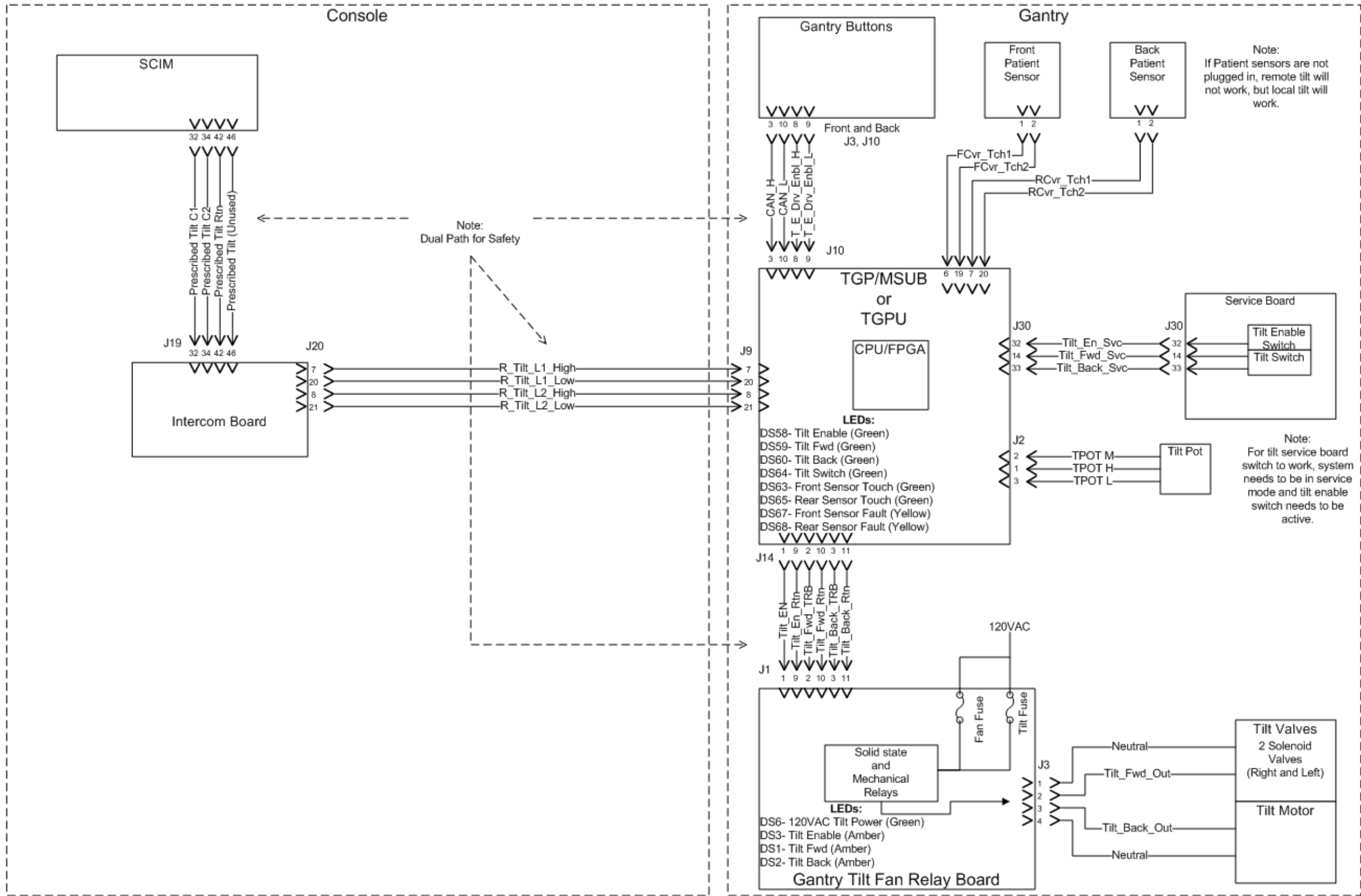


Figure 7-2 2399727 Sheet 1, Rev. 1

Section 3.0 — Table Cradle Sync - 2399728

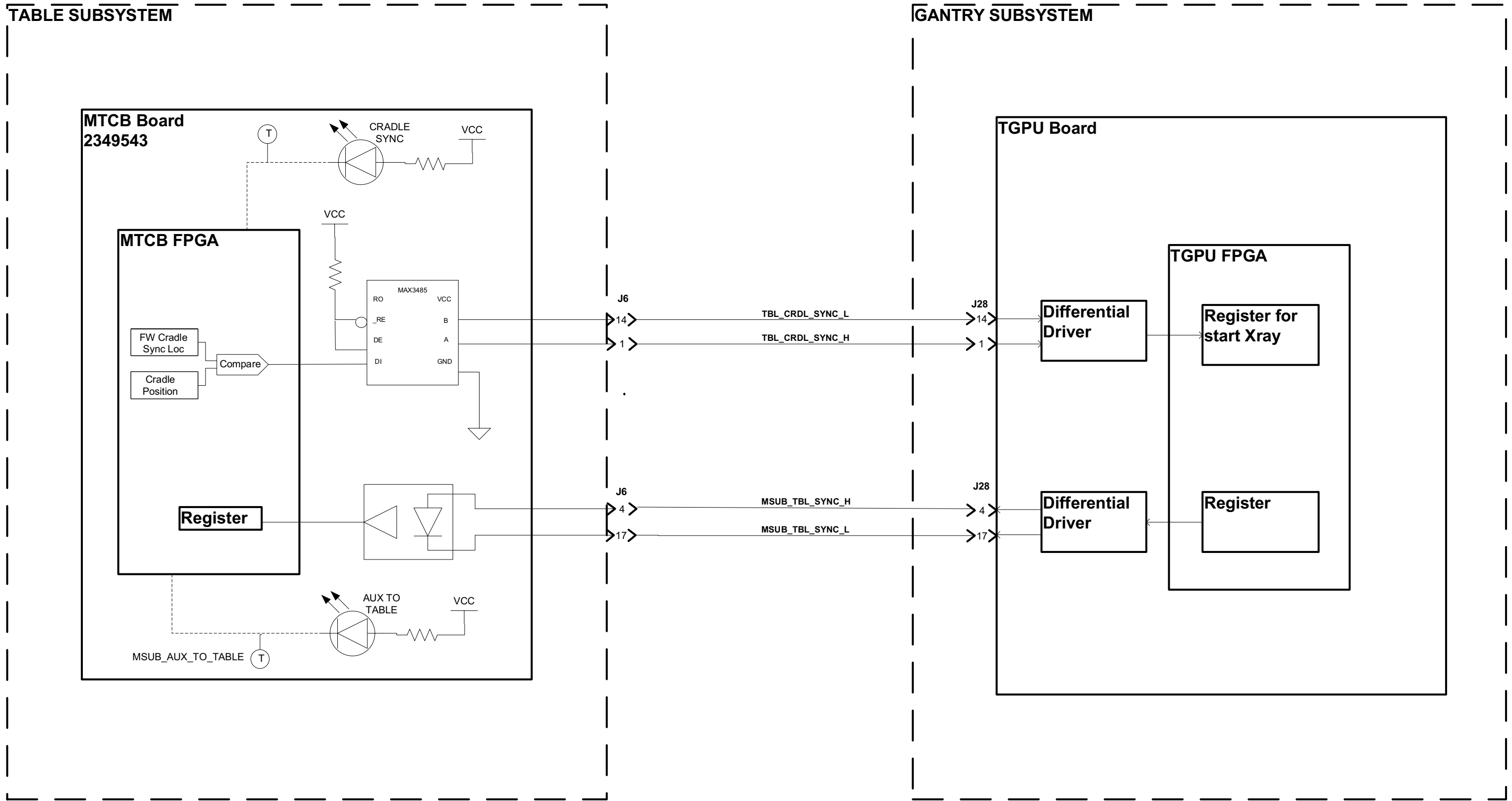


Figure 7-3 2399728 Sheet 1, Rev. 1

Section 4.0 — Table Cradle Drive - 5127766

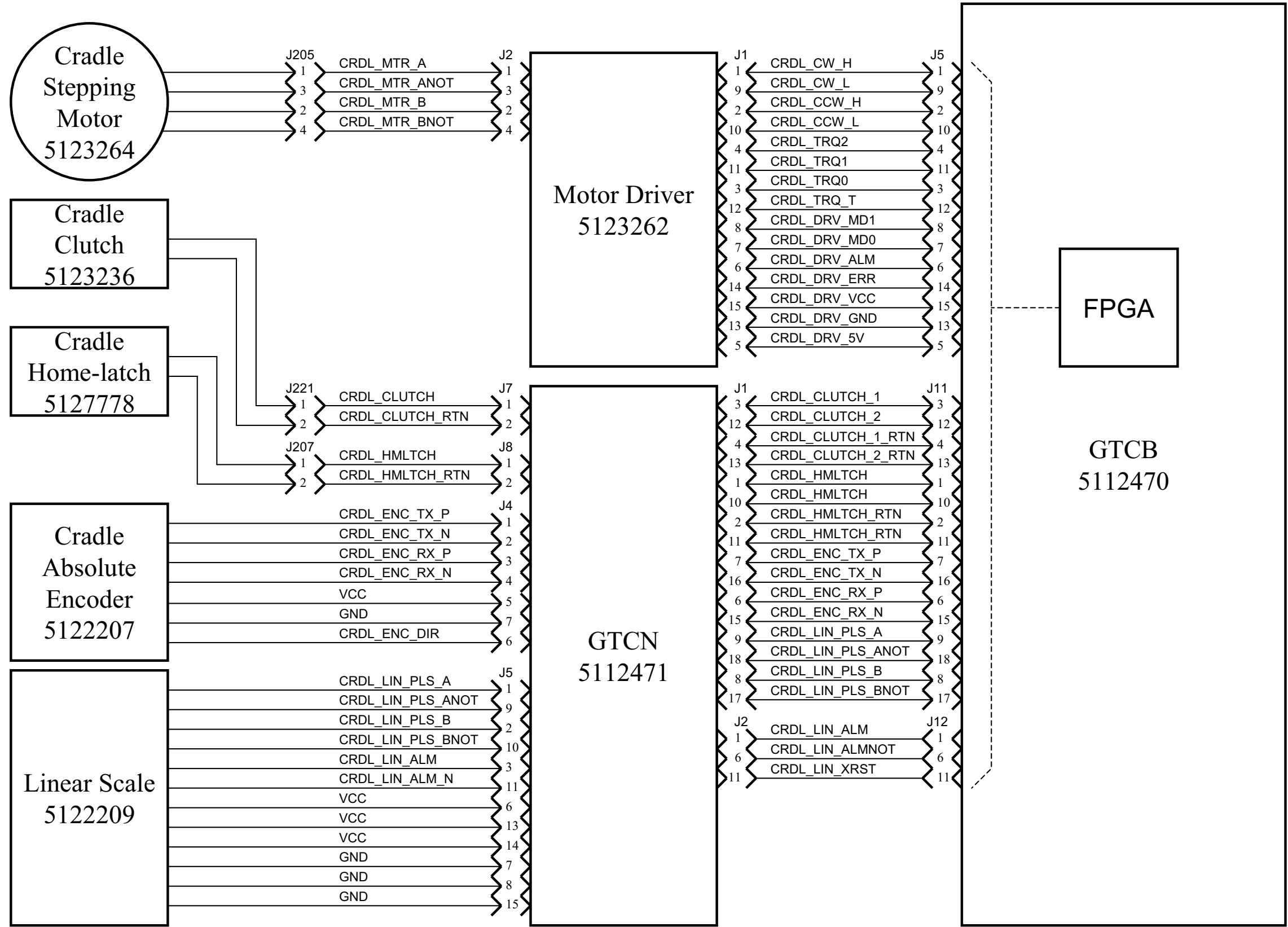


Figure 7-4 5127766 Sheet 1, Rev. 1

Section 5.0 — Table Cradle Intermediate Support Drive - 5127766

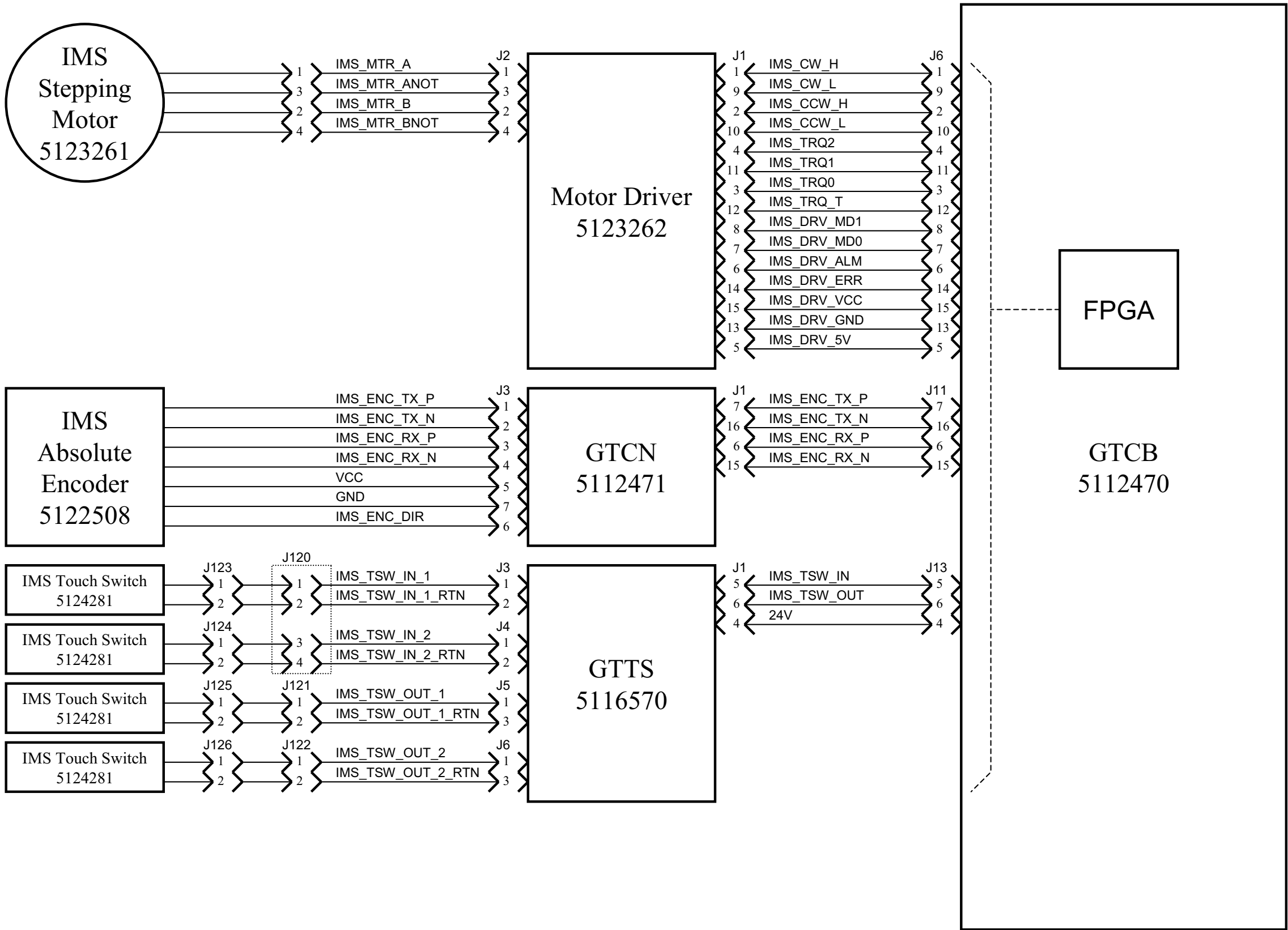


Figure 7-5 5127766 Sheet 2, Rev. 1

Section 6.0 — Table Elevation Drive - 5127766

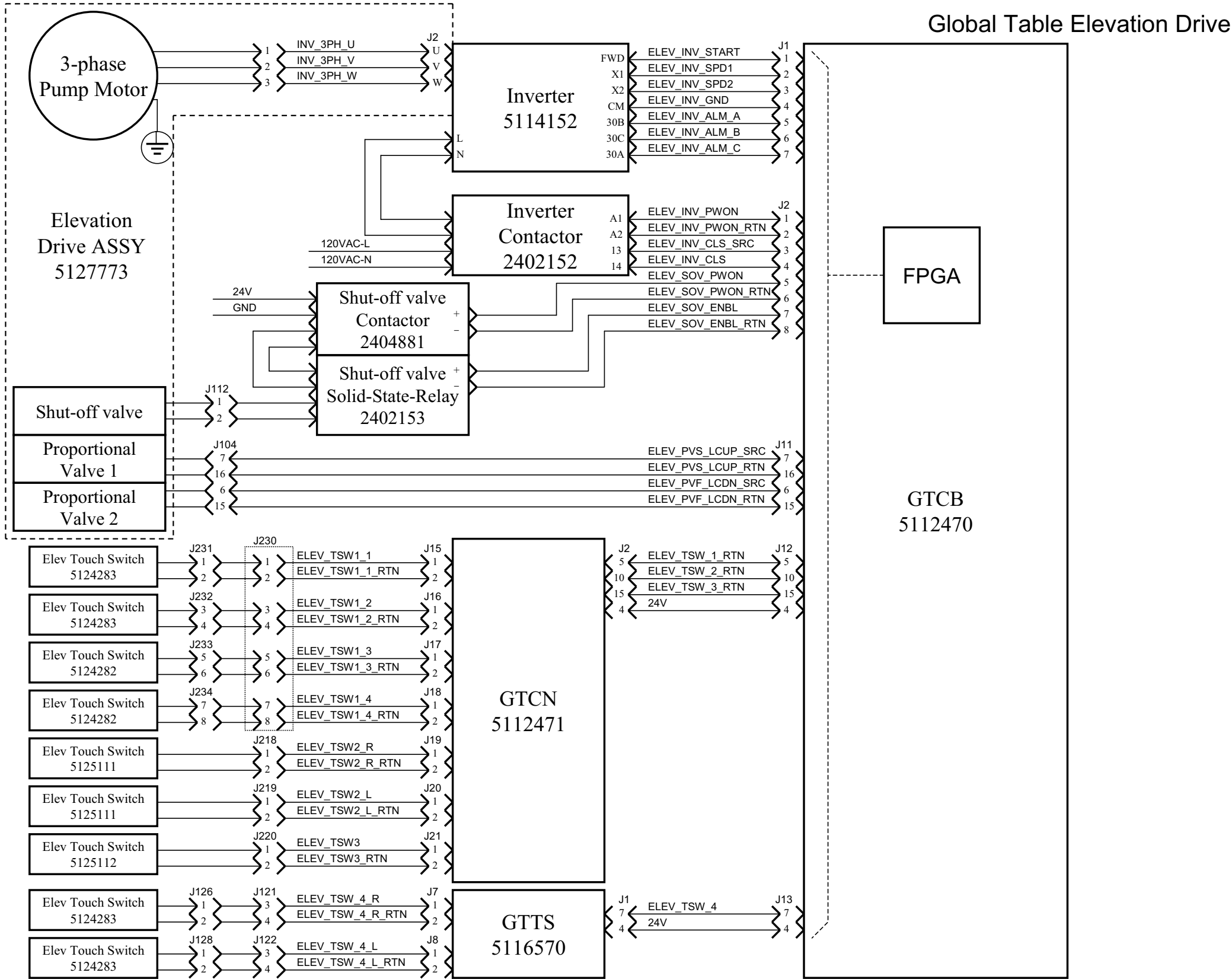


Figure 7-6 5127766 Sheet 3, Rev. 1

OVER VIEW



Page 52

Section 8.0 — Table Touch Switches - 5121647

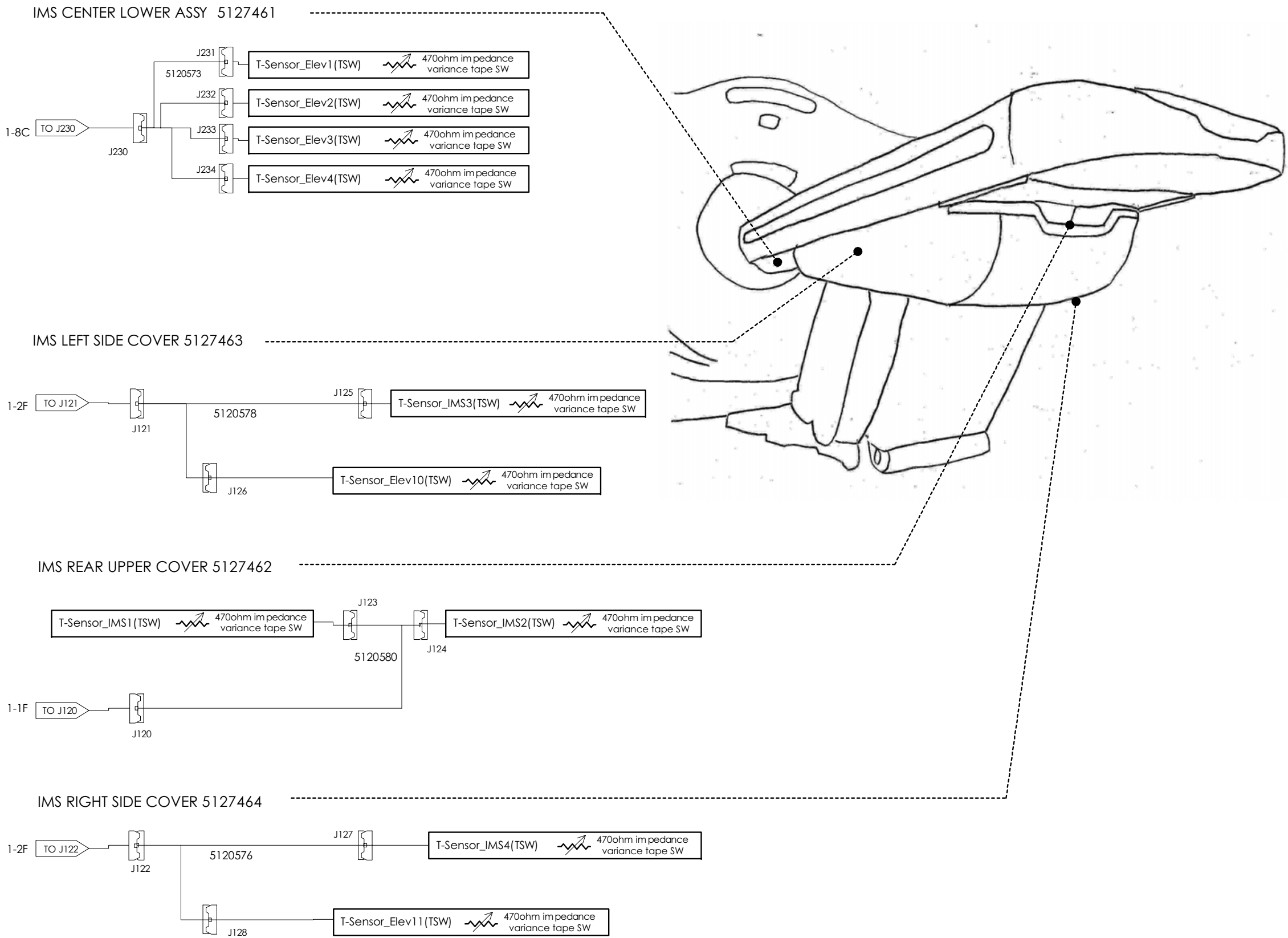
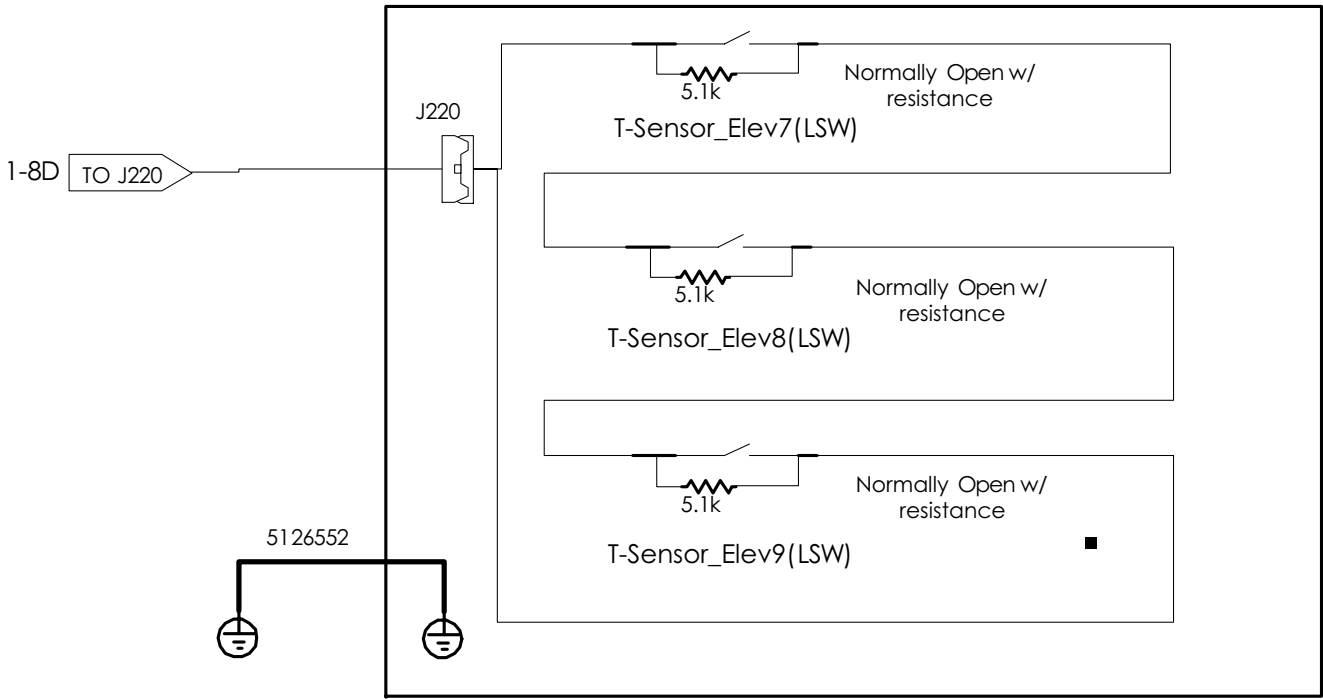
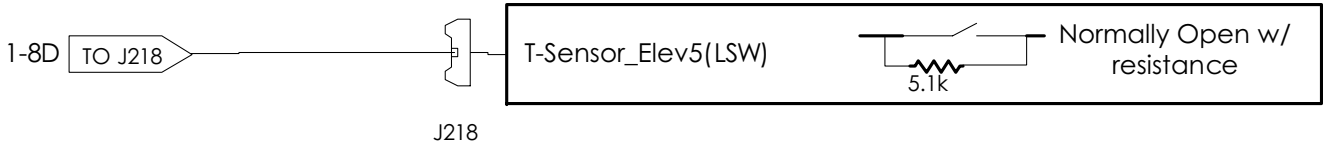


Figure 7-8 5121647 Sheet 2, Rev 3

GT REAR TOUCH SWITCH 5125112



TOUCH SENSOR R ASSY 5127469



TOUCH SENSOR L ASSY 5127470

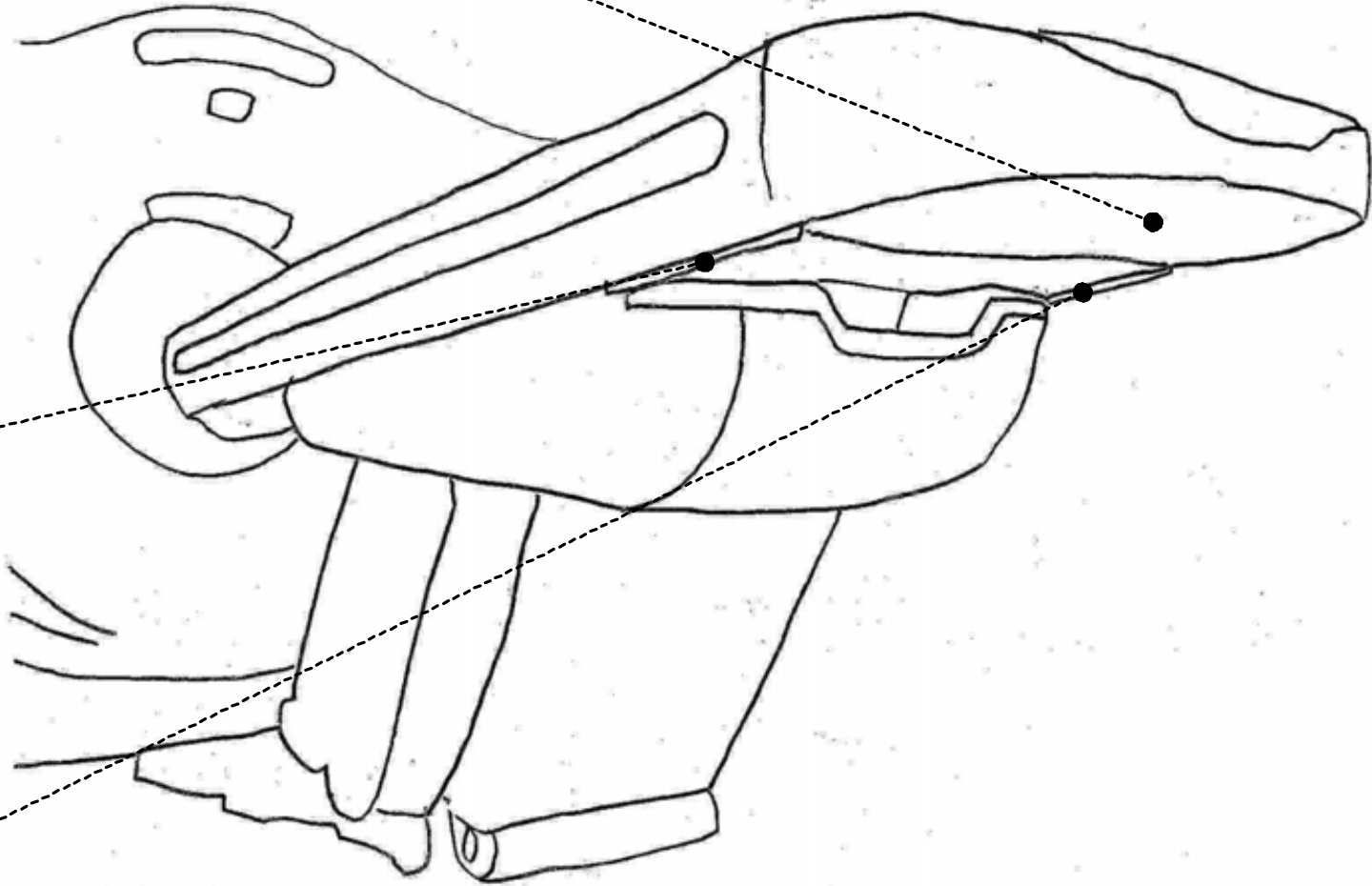
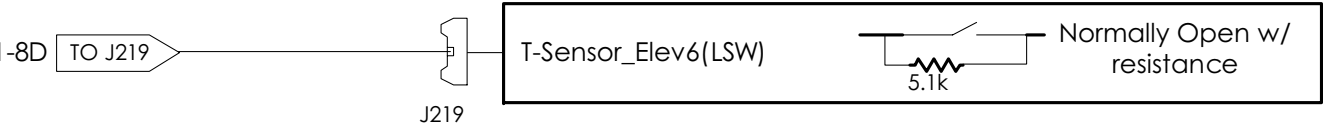


Figure 7-9 5121647 Sheet 3, Rev 3

Section 9.0 — Table Footswitch - 5121647

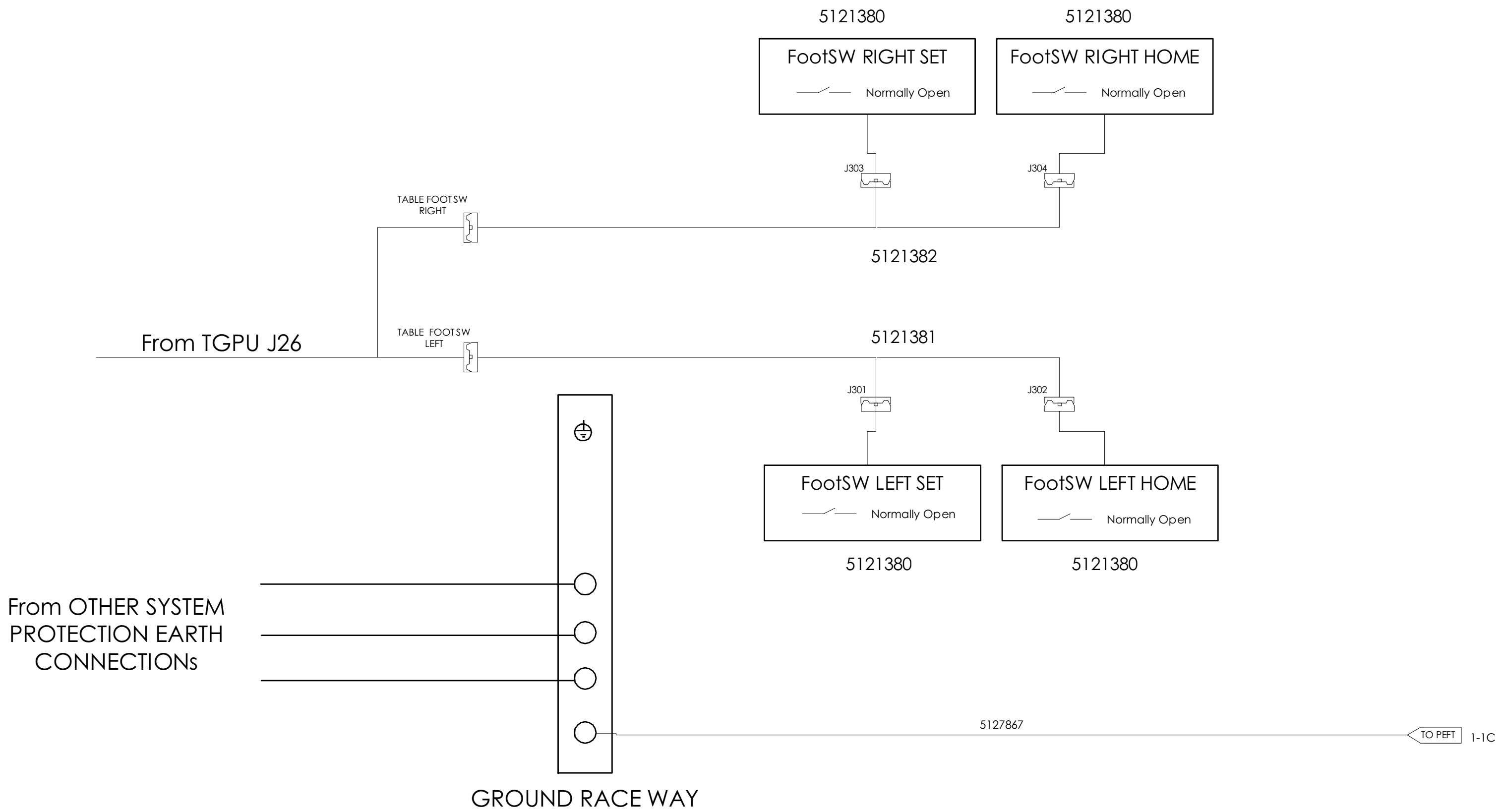


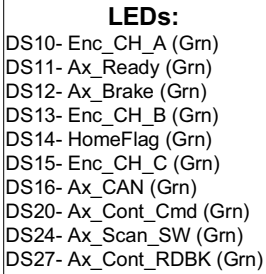
Figure 7-10 5121647 Sheet 4, Rev 3

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Chapter 8

Axial Control

Stationary



Section 1.0 —Axial Servo Control / Dynamic Brake - 2399731

Section 2.0 — Axial Power Distribution - 2399732

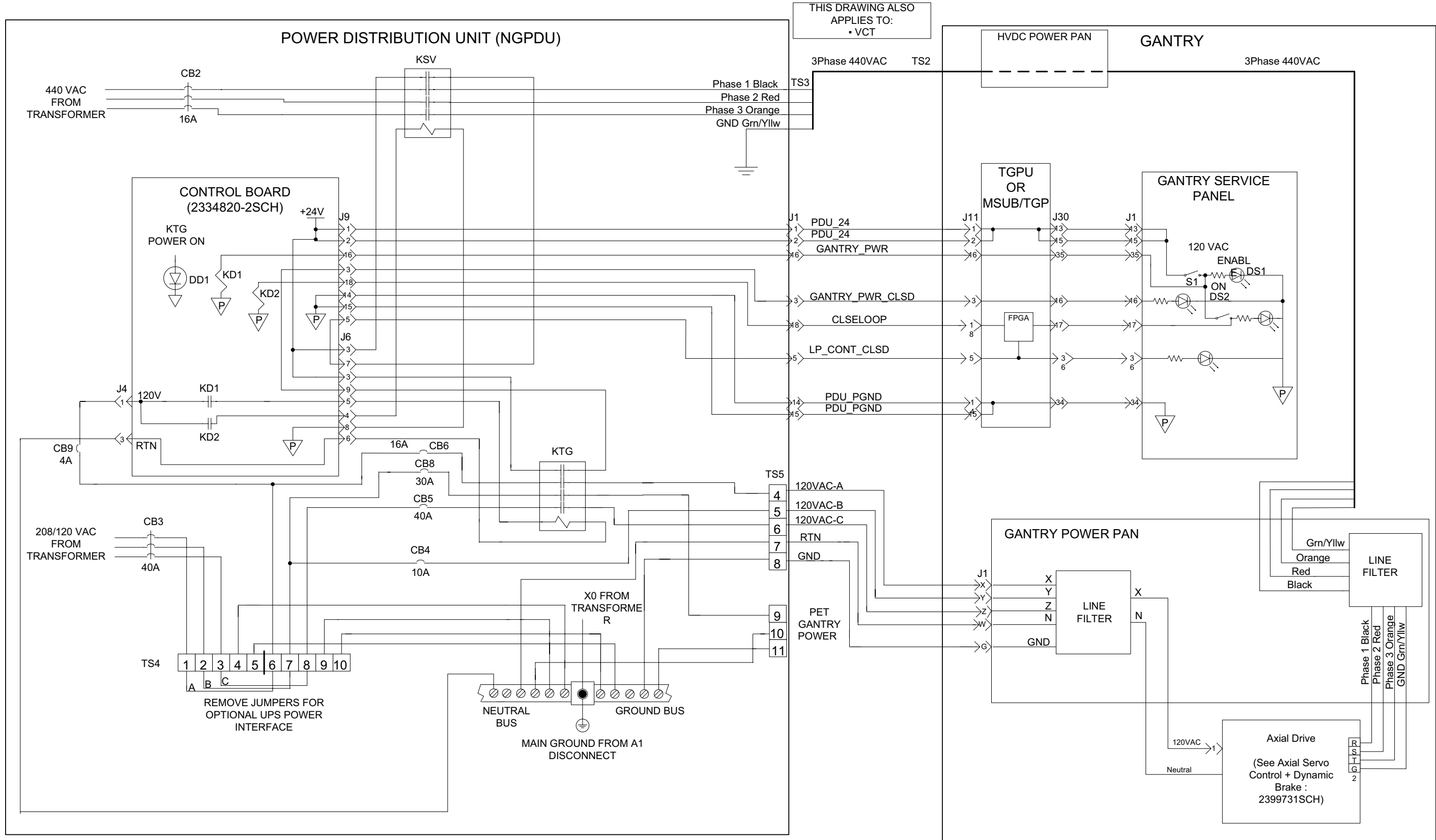


Figure 8-2 2399732 Sheet 1 Rev. 2

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Chapter 9

Operator/Patient I/O

Section 1.0 — Gantry Operator Interface and Displays - 2399733

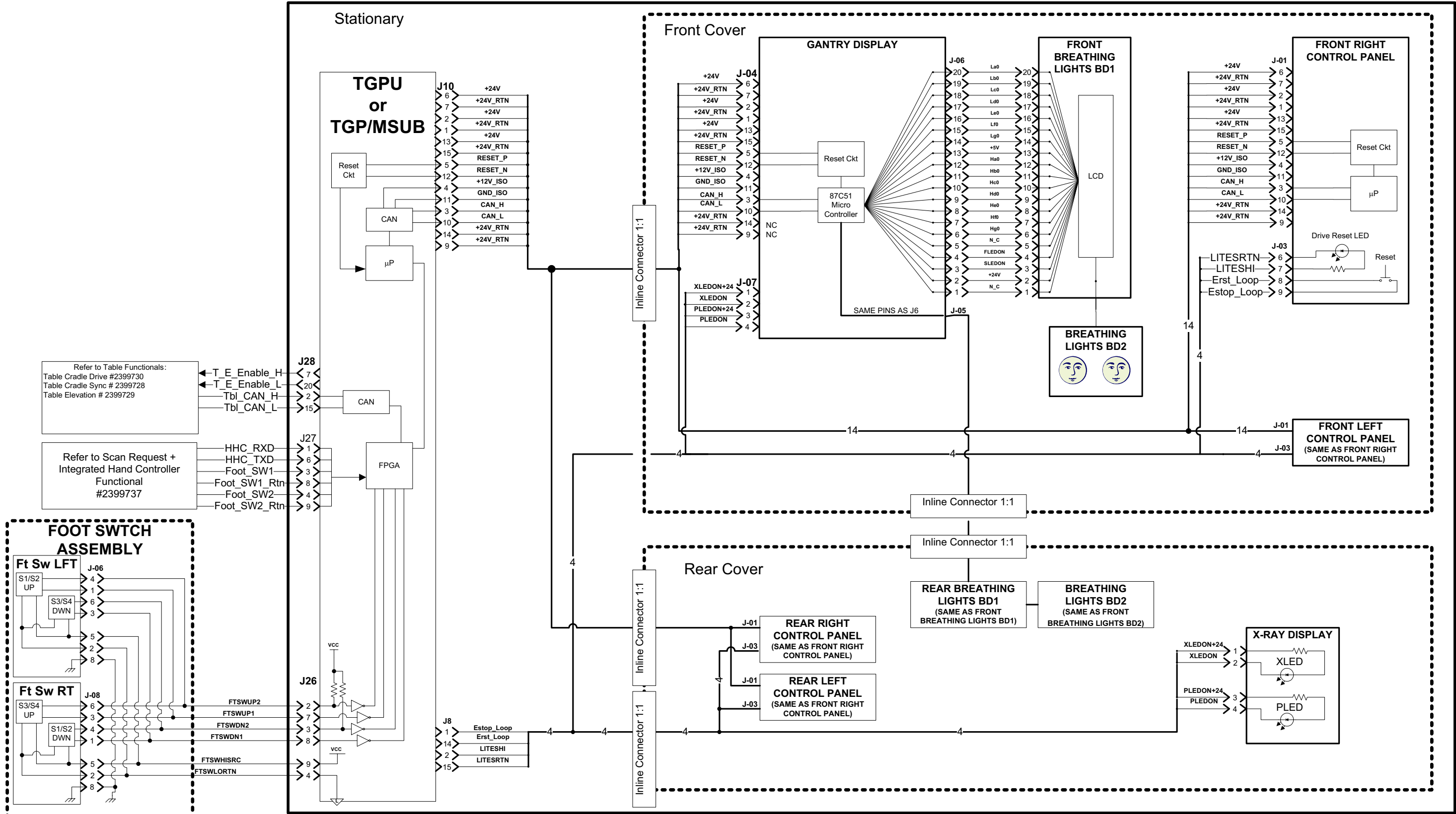


Figure 9-1 2399733 Sheet 1, Rev. 2

Section 2.0 — Gantry Reset from Console - 2399734

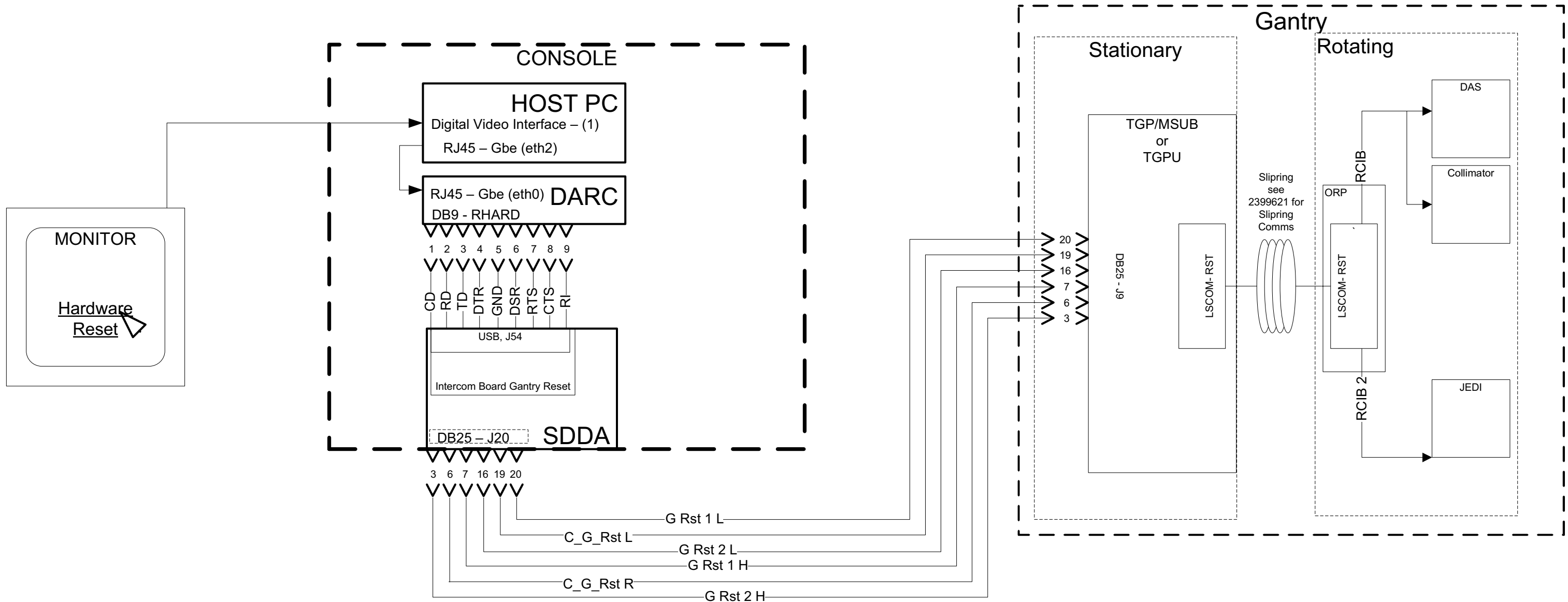


Figure 9-2 2399734 Sheet 1, Rev. 2

Section 3.0 — Cardiac Options - 2399735

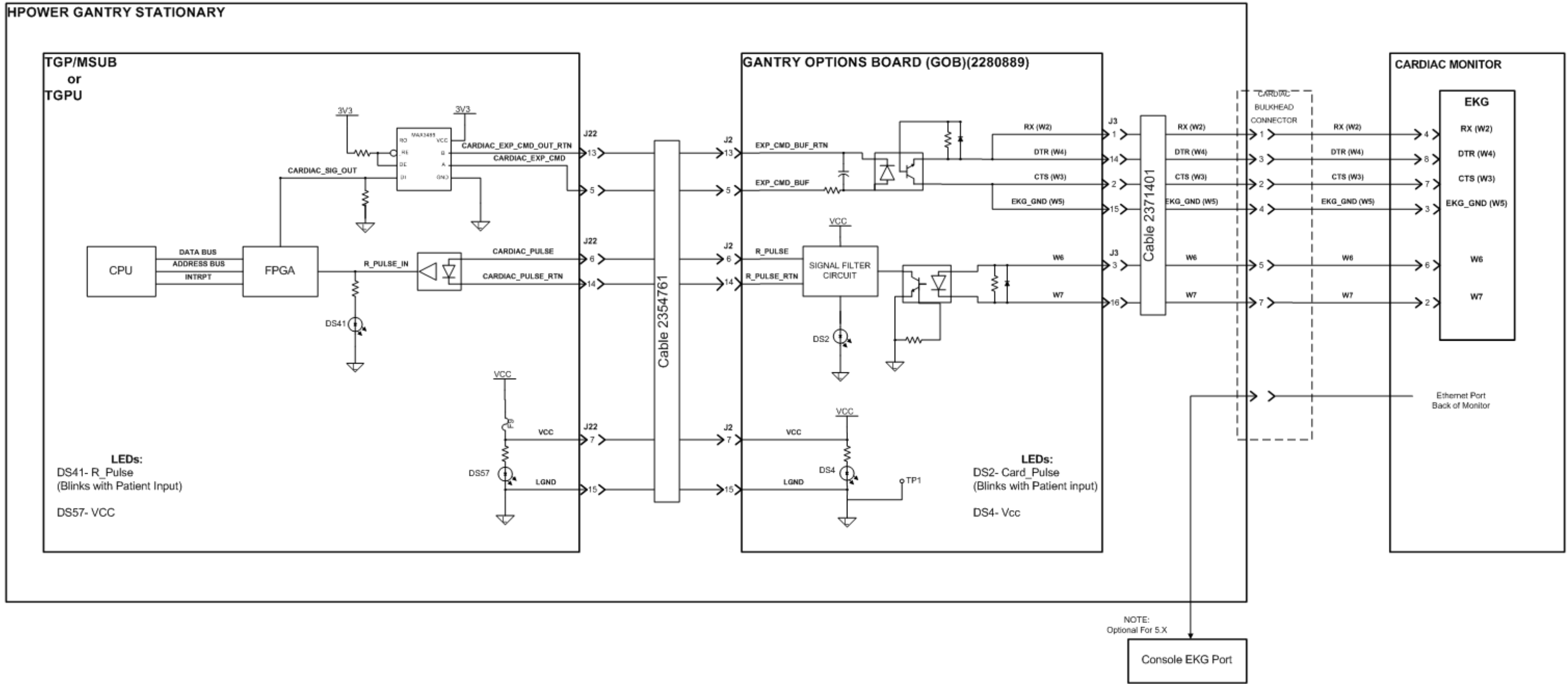


Figure 9-3 2399735 Sheet 1, Rev. 1

Section 4.0 — Pulmonary Options - 2399736

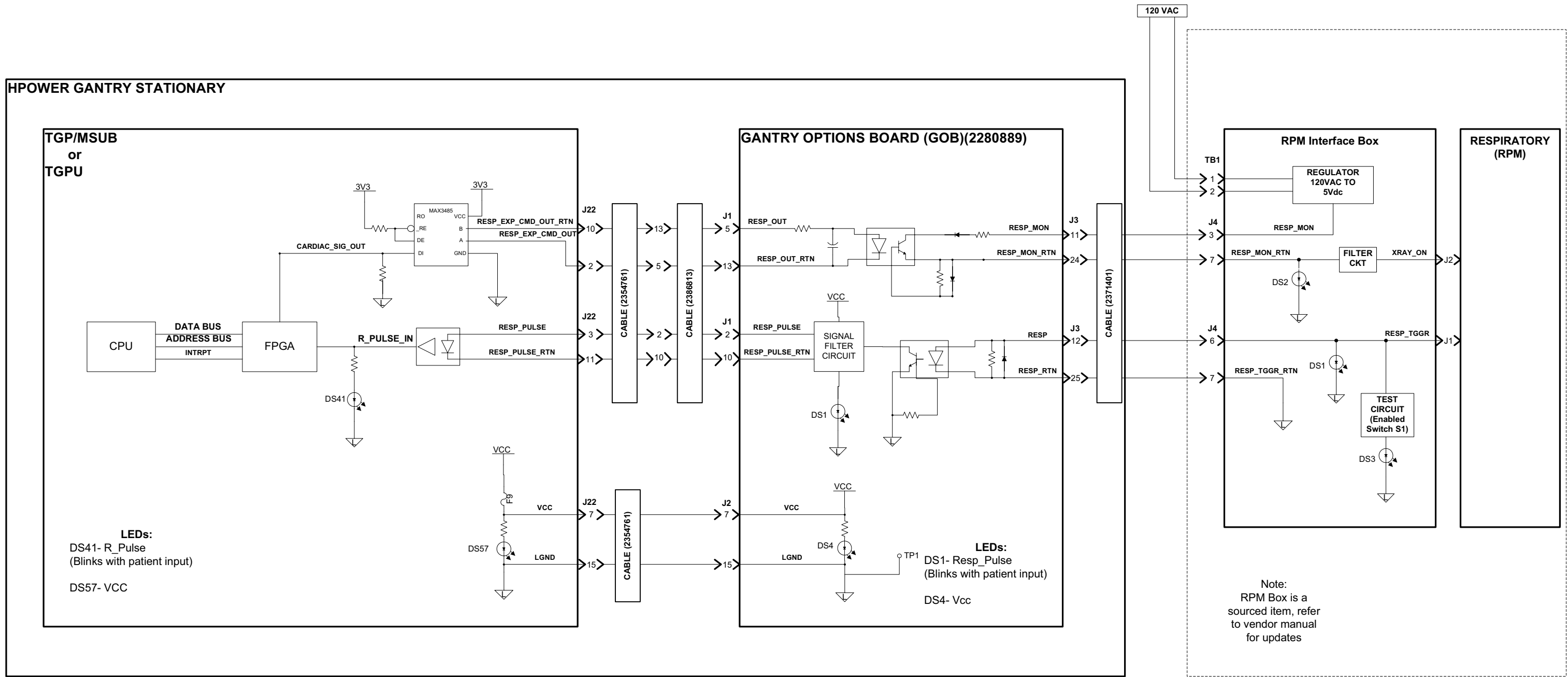


Figure 9-4 2399736 Sheet 1, Rev. 1

Section 5.0 — Scan Footswitch and IHHC Controller - 2399737

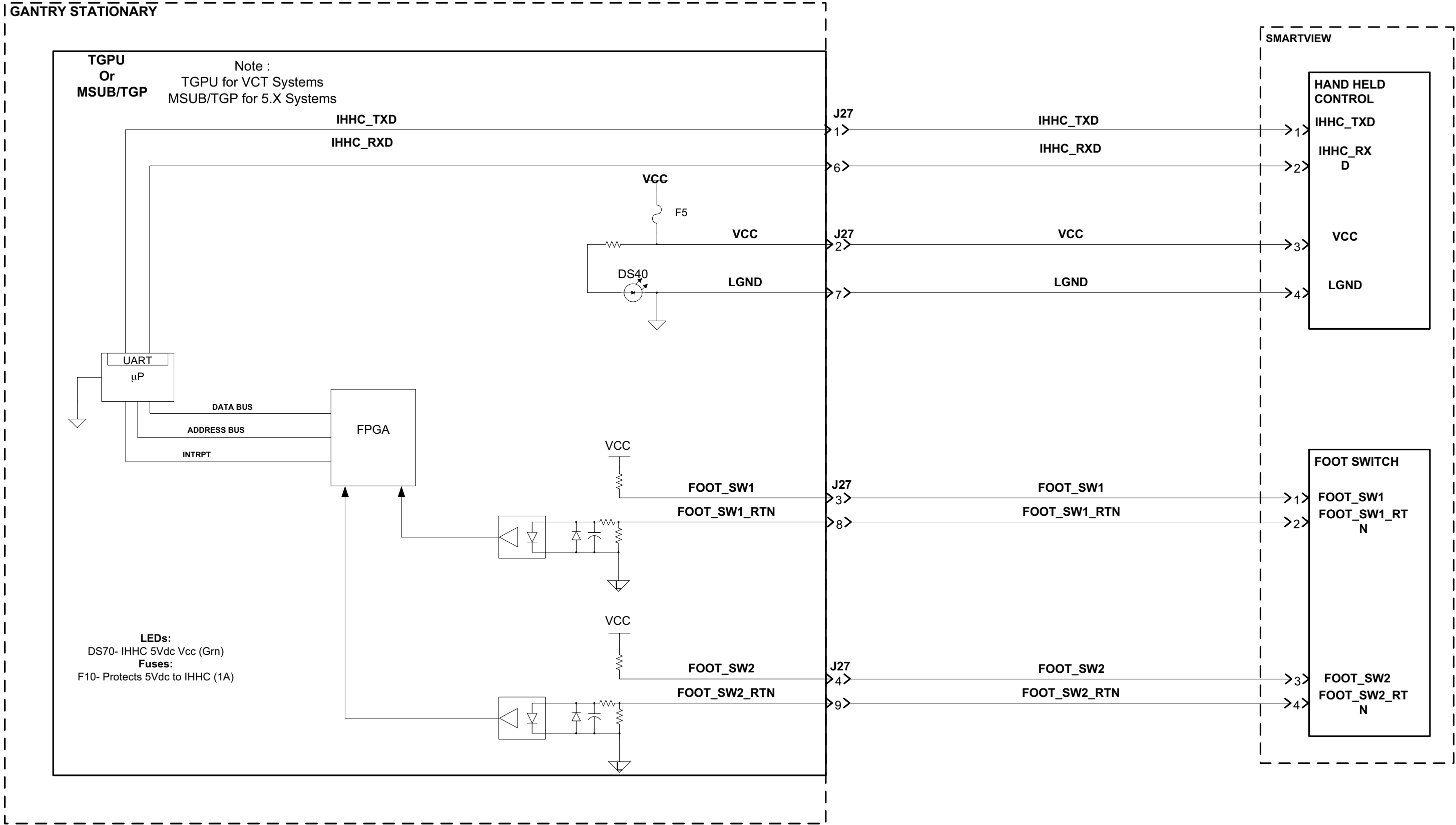


Figure 9-5 2399737 Sheet 1, Rev. 2

Section 6.0 — Site X-Ray Light, System Light, Door Interlock - 2399738

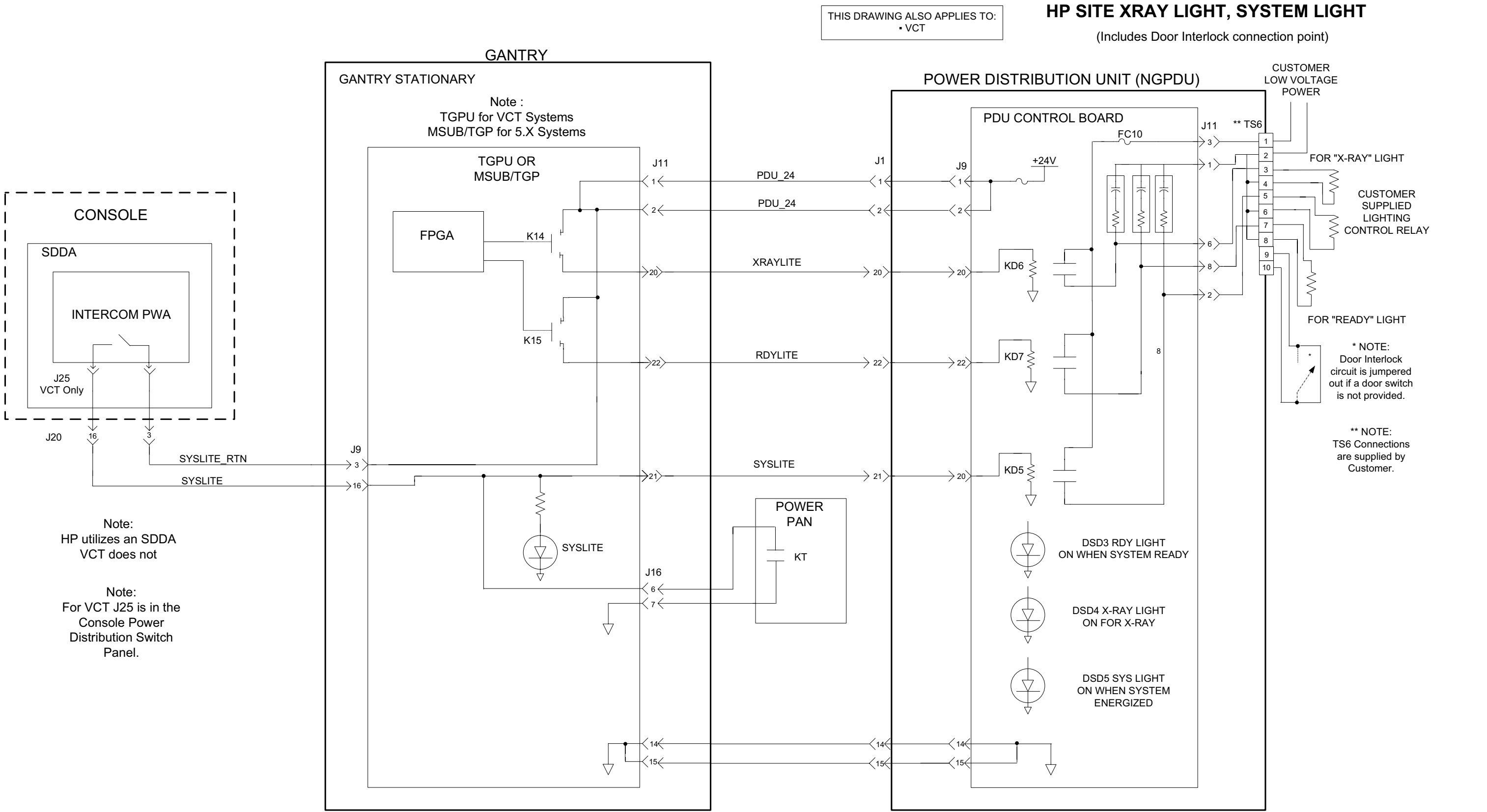


Figure 9-6 2399738 Sheet 1, Rev. 3

Section 7.0 — Autovoice / Intercom - 2399739

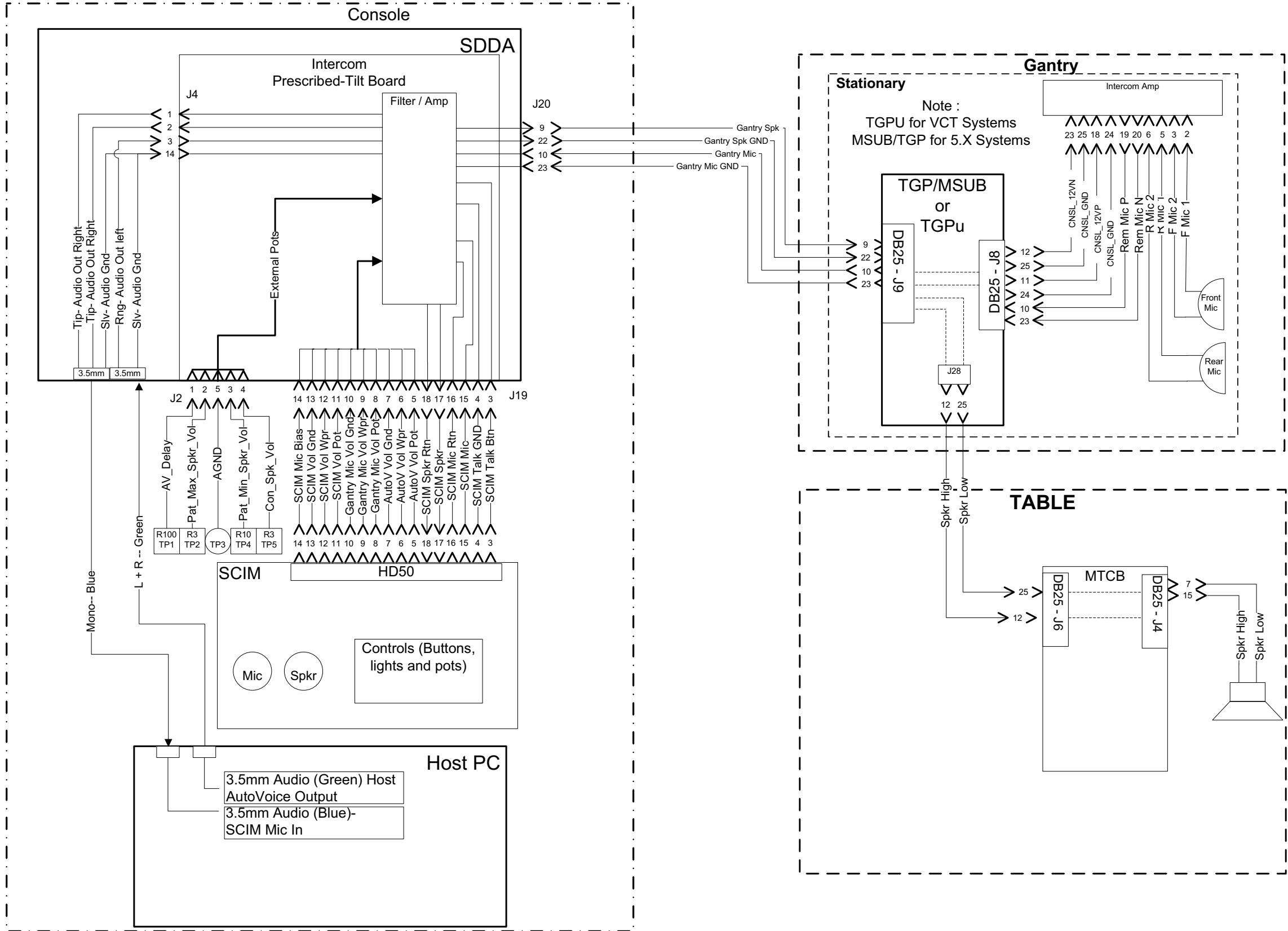


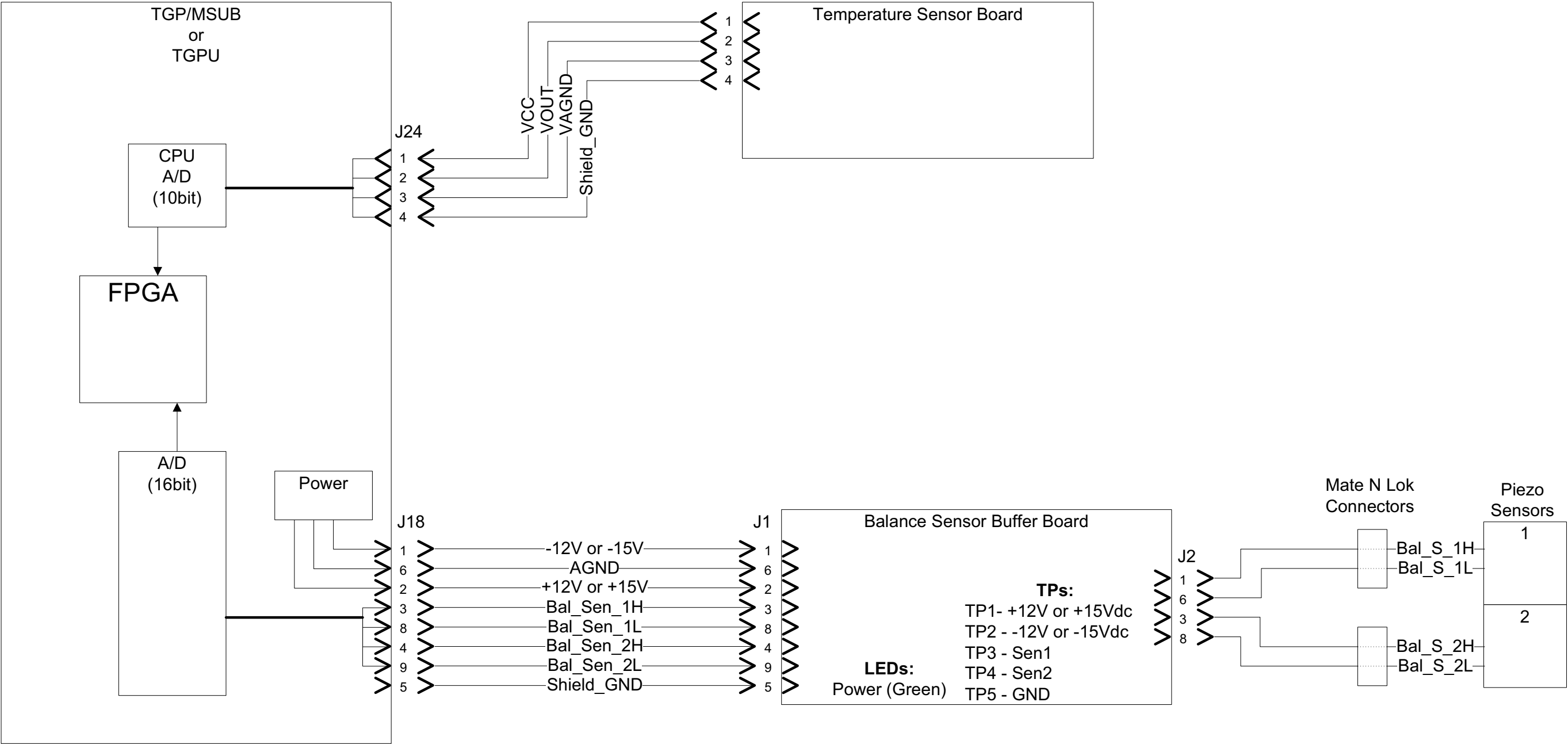
Figure 9-7 2399739 Sheet 1, Rev. 3

Chapter 10

System Monitoring

Section 1.0 — Gantry Temperature and Balance Sensors - 2399740

Note :
TGPU for VCT Systems
MSUB/TGP for 5.X Systems



Note:
MSUB powered by +-15Vdc
TGPU powered by +-12Vdc

Figure 10-1 2399740 Sheet 1, Rev. 3

Section 2.0 — Stationary Gantry Fans & Air Temperature Control - 5128040

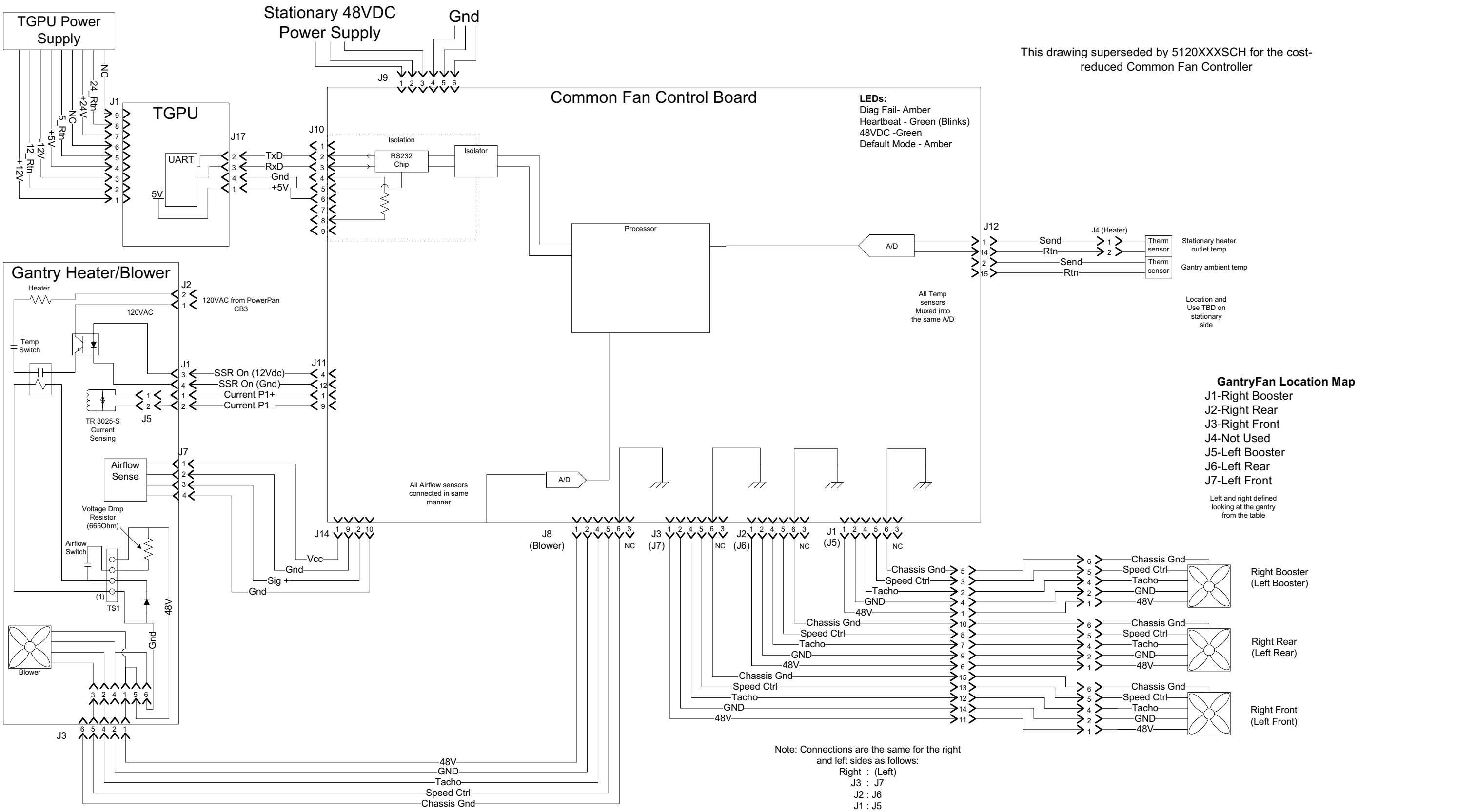
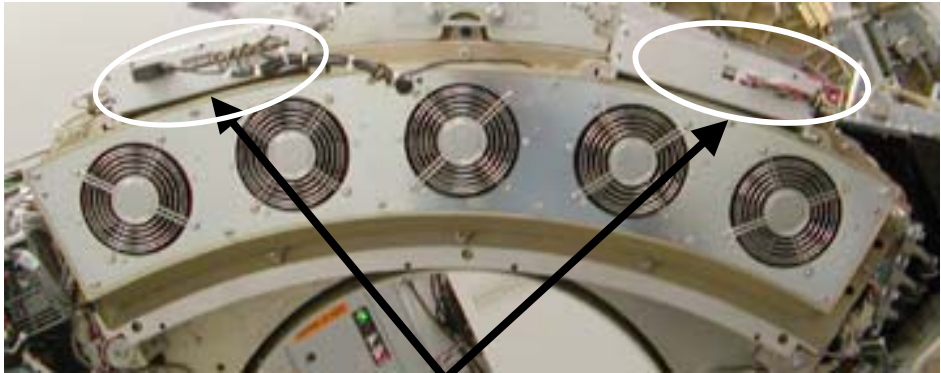


Figure 10-2 5128040 Sheet 1, Rev. 1

Section 3.0 — Detector Heater Control (2005 Style)



2005 Style: Molex connector style for most CFC and DHC cables

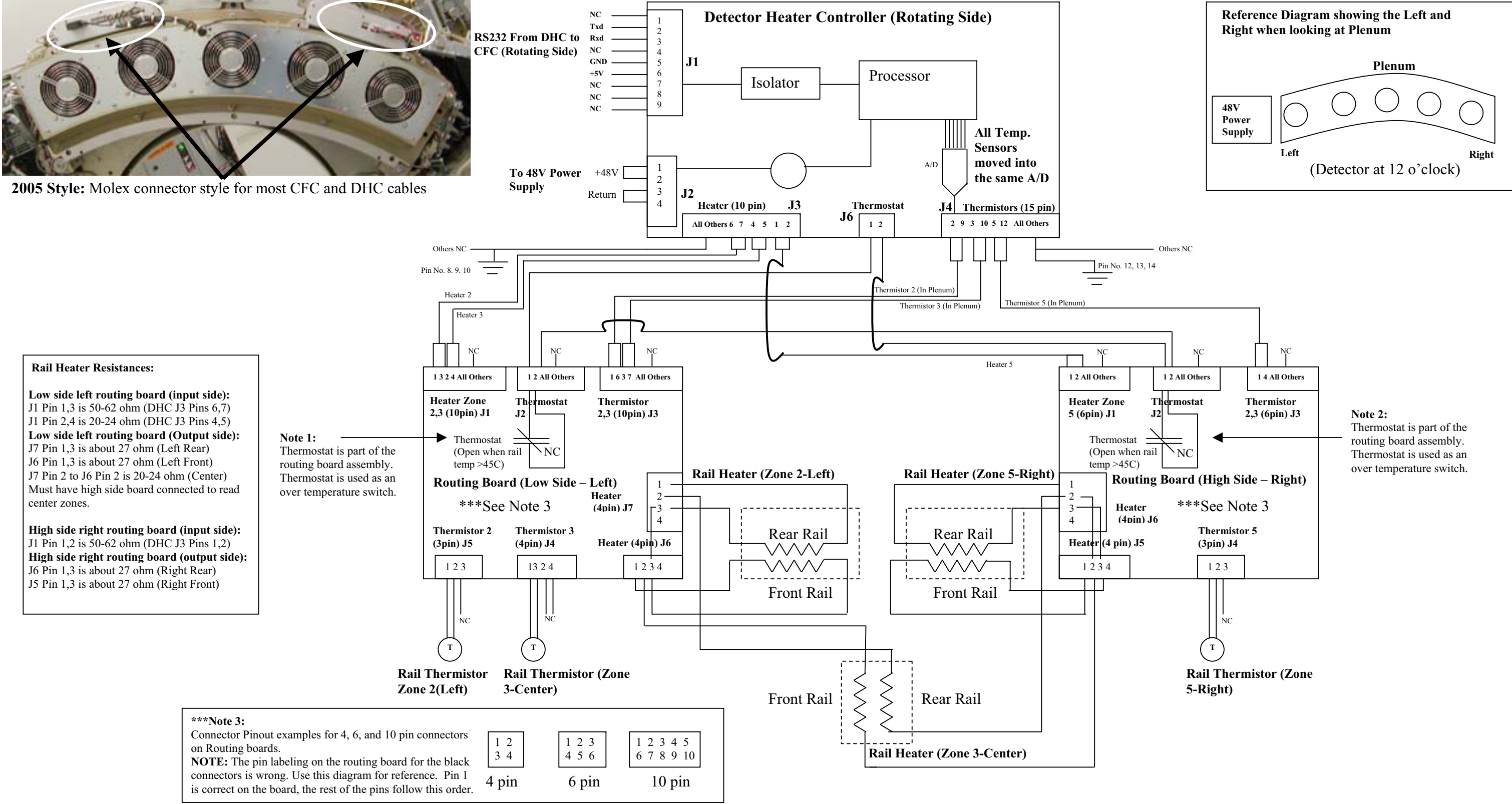


Figure 10-3 Detector Heater Control (2005 Style) Functional Diagram, Rev 4

Section 4.0 — Detector Heater Control (2006 Style)

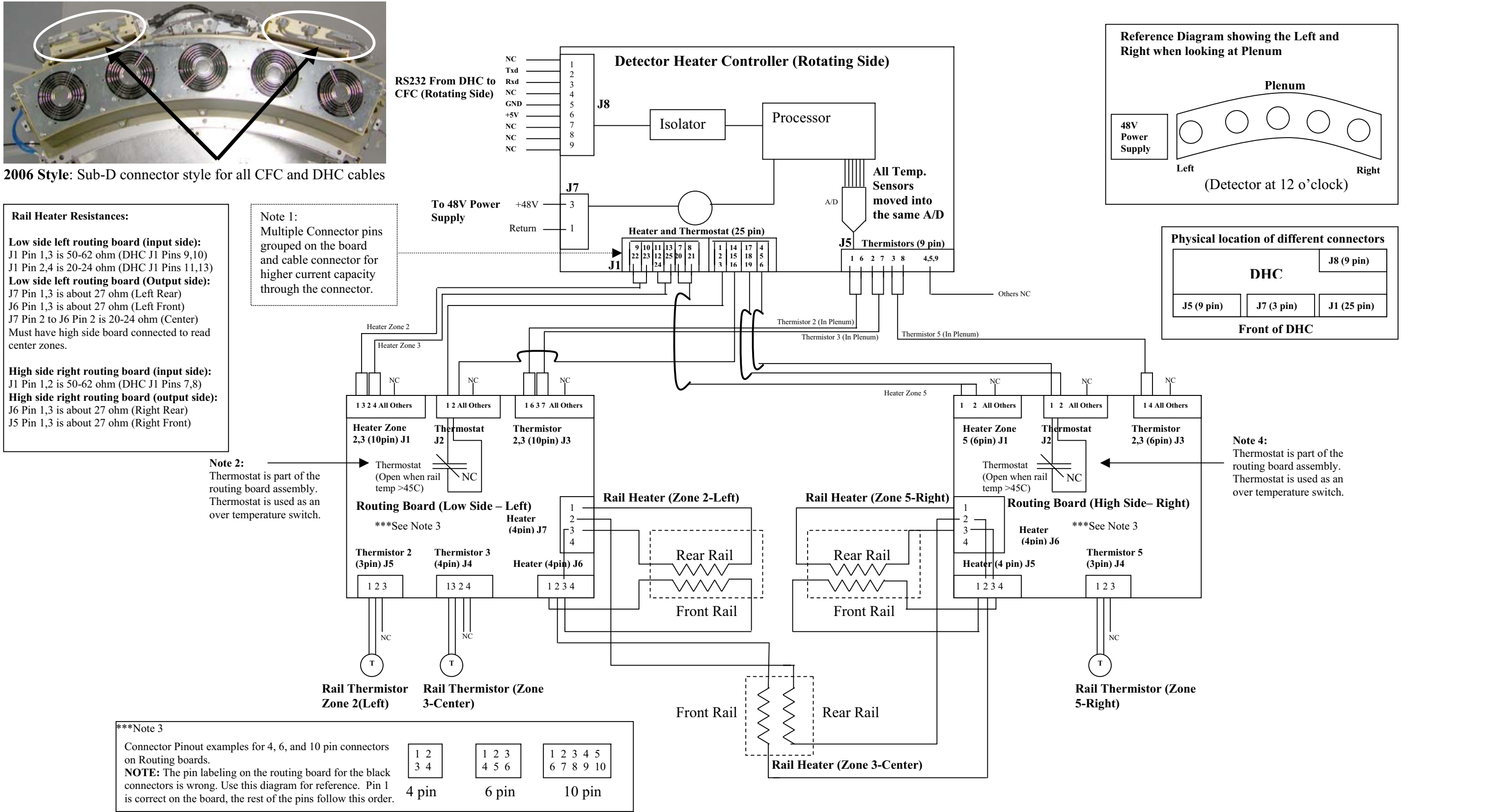


Figure 10-4 Detector Heater Control (2006 Style) Functional Diagram, Rev 1

Section 5.0 — Detector Plenum Fan Control (2005 Style)

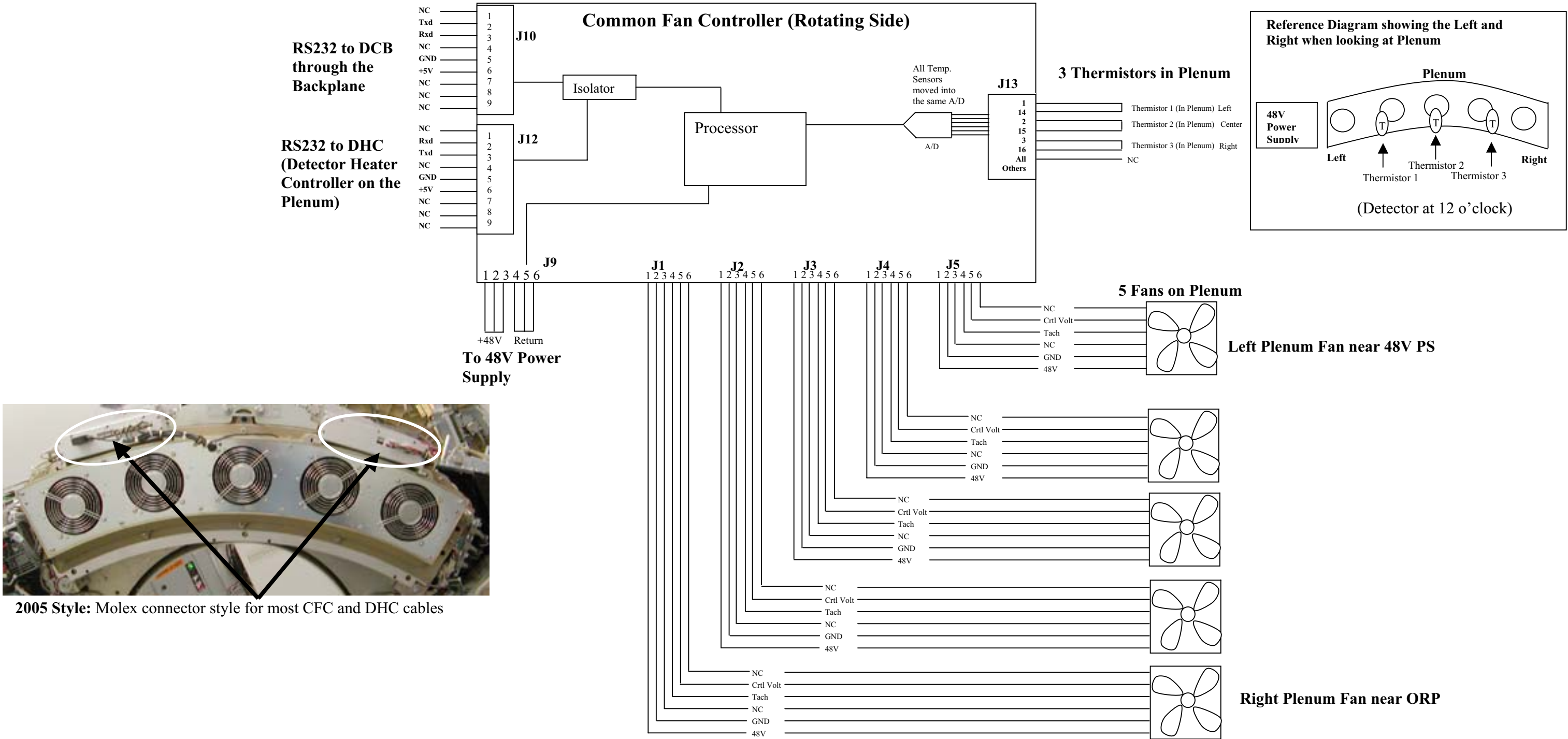


Figure 10-5 Detector Plenum Fan Control (2005 Style) Functional Diagram, Rev 3

Section 6.0 — Detector Plenum Fan Control (2006 Style)

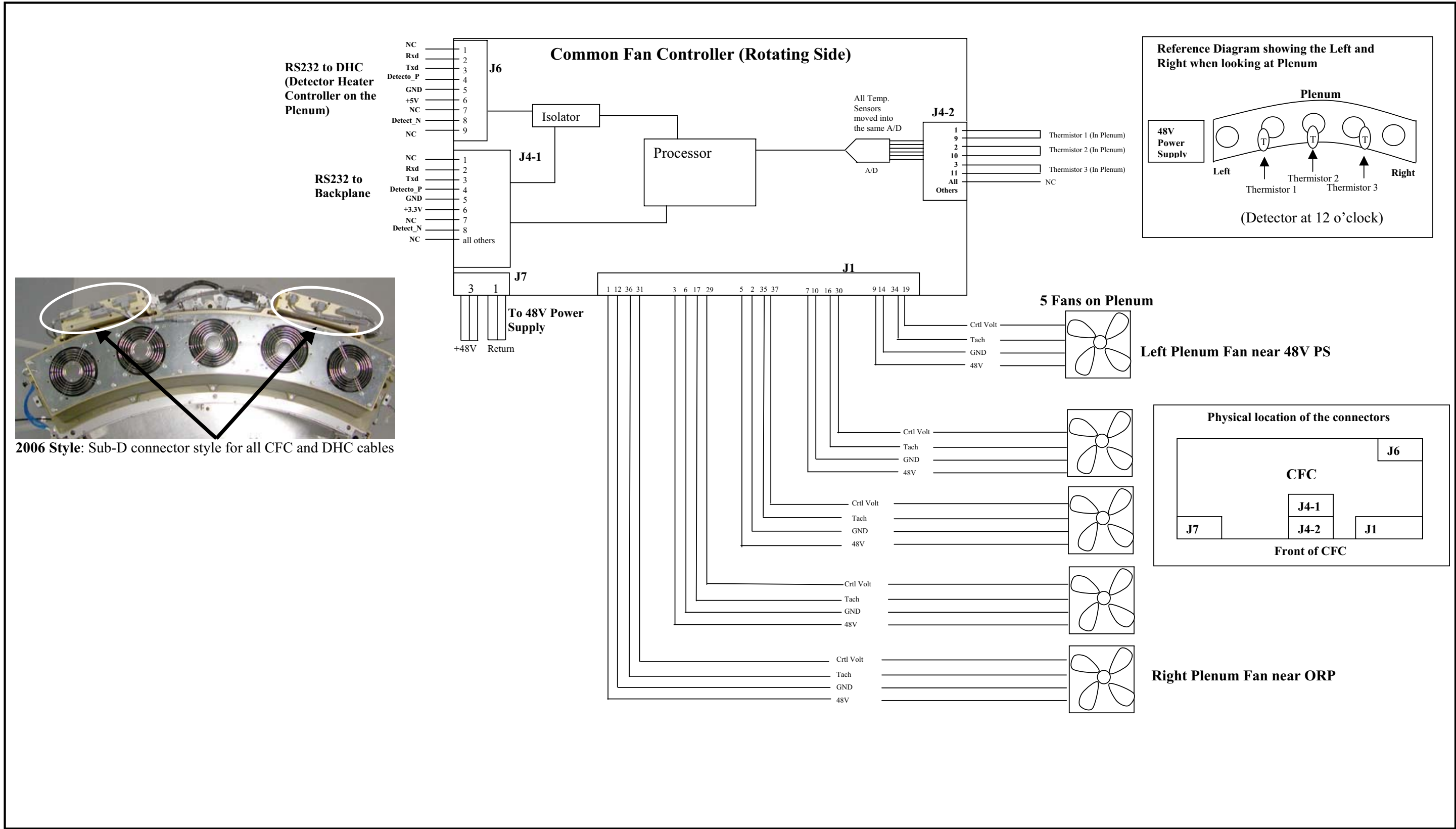


Figure 10-6 Detector Plenum Fan Control (2006 Style) Functional Diagram, Rev 1

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Chapter 11

System Power Control

Section 1.0 — System E-Stop and Power Interlocks - 2399742

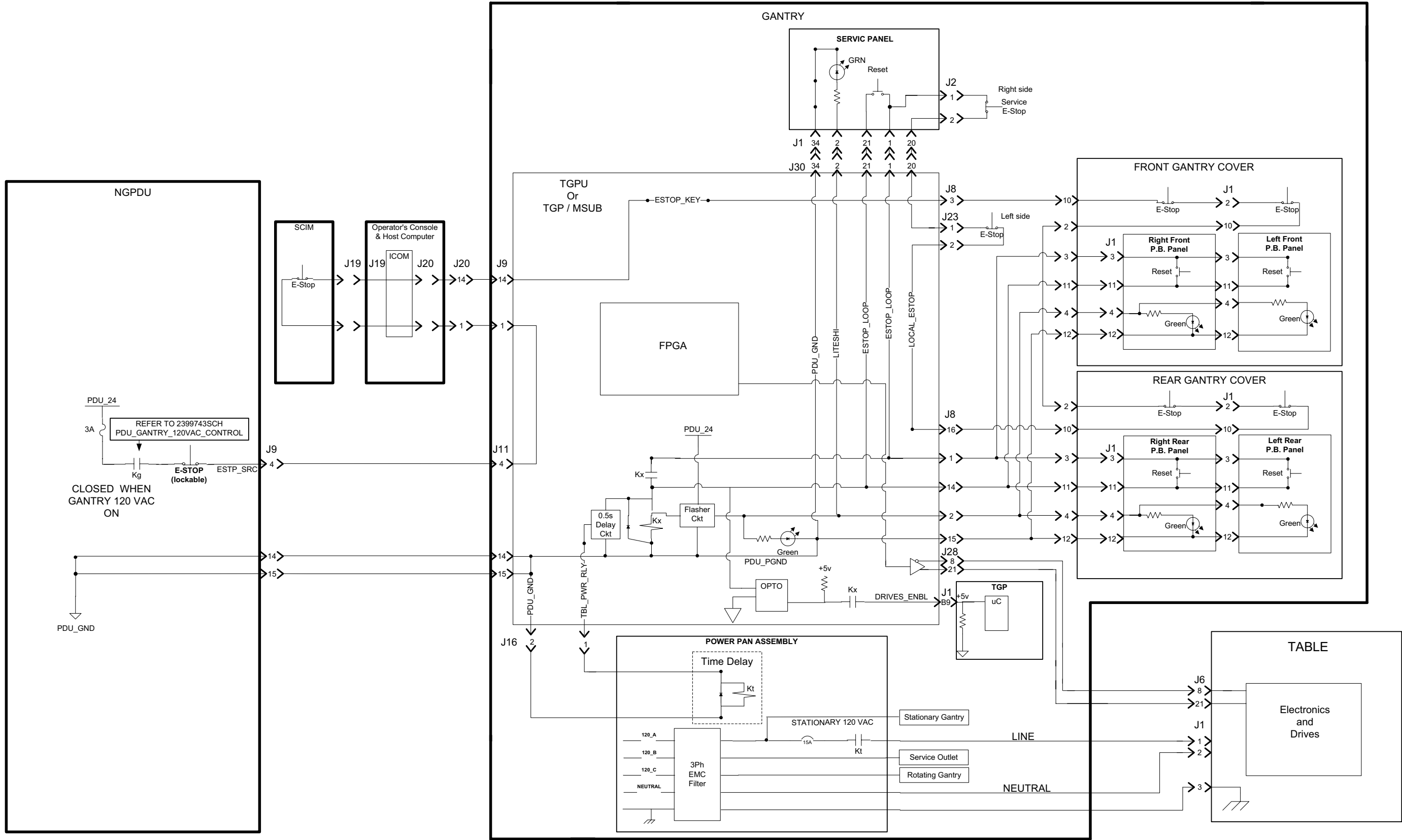


Figure 11-1 2399742 Sheet 1, Rev. 3

Section 2.0 — HVDC/120V Control - 2399744

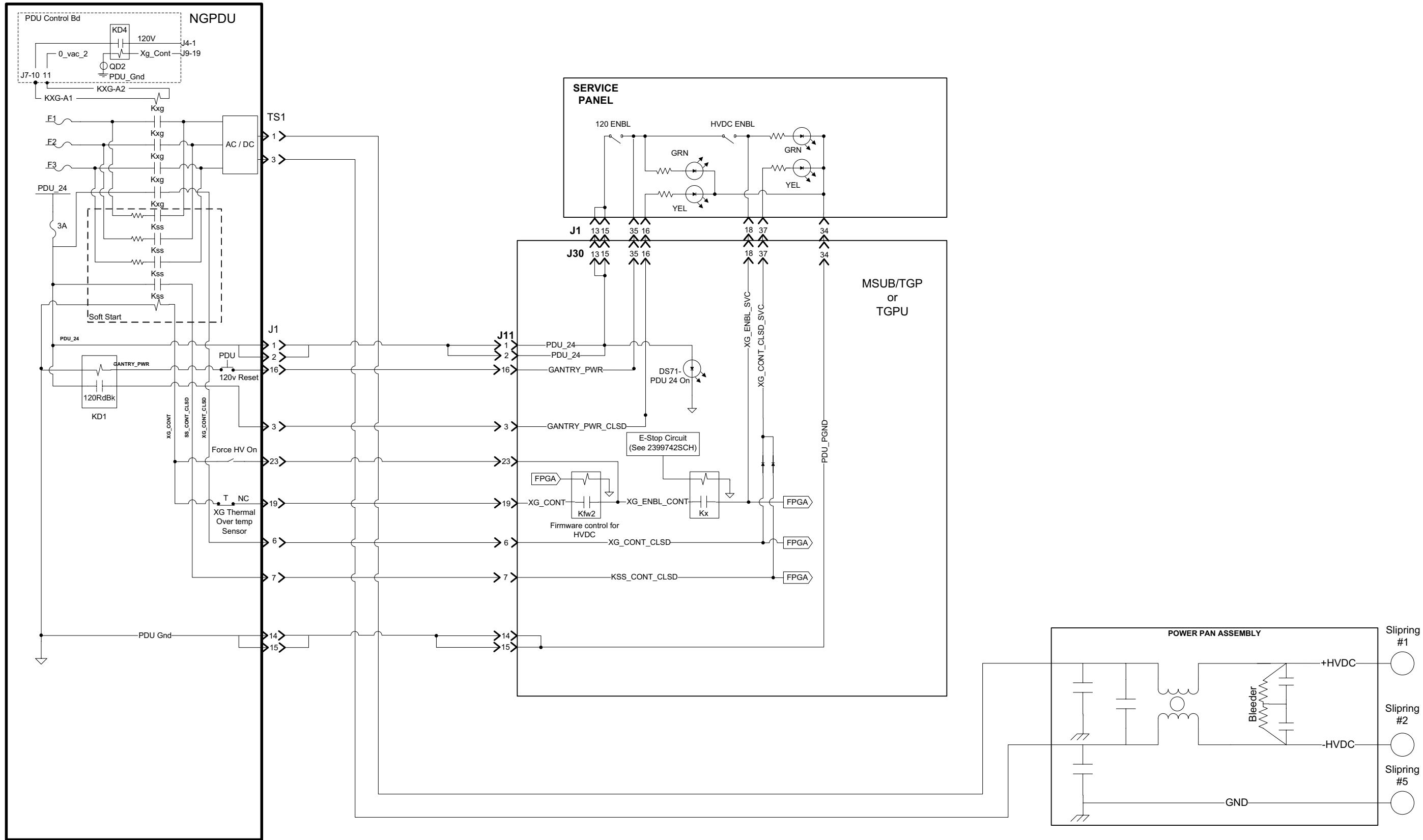


Figure 11-2 2399744 Sheet 1, Rev 2

Section 3.0 — Gantry Power Distribution - 5128042

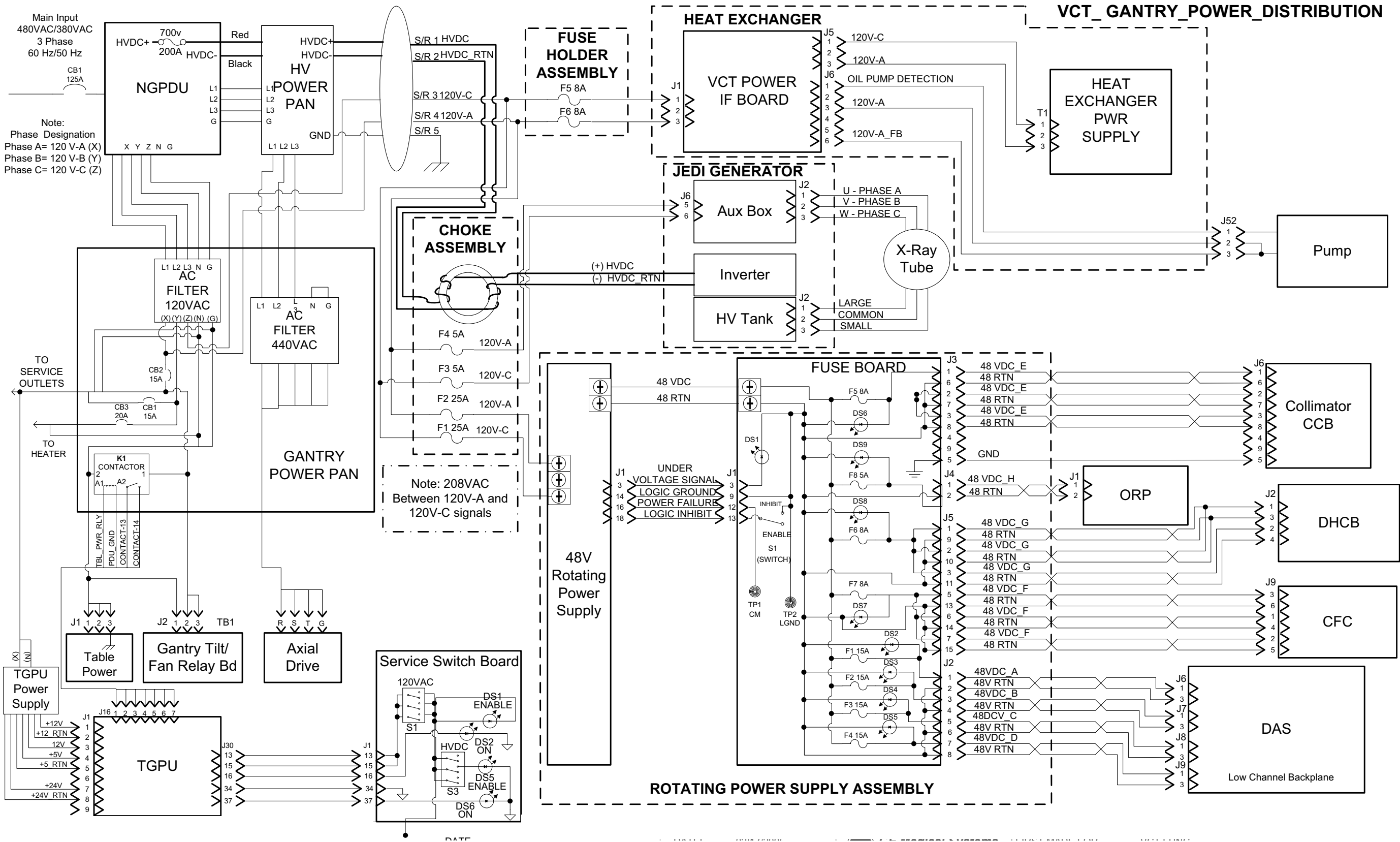


Figure 11-3 5128042 Sheet 1, Rev 2

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